Let's talk about Thurstone & Co.: An information-theoretical analysis workflow for comparative judgments

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Abstract

This study revisits Thurstone's law of comparative judgment (CJ) by examining three prominent issues from the CJ literature. First, it discuss the reliance on strong simplifying assumptions in the modeling of CJ data. The study argues that while such assumptions can simplify trait measurement, they may fail to capture complex traits or heterogeneous stimuli, leading to biased or unreliable trait estimates. Second, it highlights the limited attention given to relevant CJ assessment design features. Here, the study contends that overlooking elements defining the CJ data-generating process can obscure their role within the assessment system and may reduce the accuracy and precision of trait estimates and statistical inferences. Third, the study identifies a disconnect between trait measurement and statistical inference. Here it argues that, while separating these processes simplifies analysis, this practice prevents the propagation of uncertainty from estimation to inference, thereby weakening statistical conclusions.

To address these issues, the study presents an umbrella analytical workflow designed to accommodate the complexity of real CJ assessments. The approach uses Causal and Bayesian inference methods to integrate Thurstone's core theoretical principles with key CJ assessment design features, and translates these elements into a coherent statistical model for analysis of CJ data.

The study concludes that this extension supports more appropriate model specification in contemporary CJ research across education, linguistics, and psychology, while highlighting the need for CJ models that are aligned with both the underlying experimental design and the assumptions of the data-generating process.

Keywords: causal inference, directed acyclic graphs, structural causal models, bayesian inference, thurstonian model, comparative judgment, probability, statistical modeling

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1. Introduction

In *comparative judgment* (CJ) studies, judges assess a specific trait or attribute across different stimuli by performing pairwise comparisons (Thurstone, 1927a,b). Each comparison produces a binary outcome that indicates which stimulus the judge perceives as having a higher level of the trait. For example, judges may compare pairs of written texts to evaluate their relative written quality (Pollitt, 2012b; van Daal et al., 2016; Lesterhuis, 2018; Coertjens et al., 2017; Goossens and De Maeyer, 2018; Bouwer et al., 2023).

Numerous studies document the effectiveness of CJ in assessing traits and competencies over the past decade. They highlight three key strengths of the method: its reliability, its validity, and practical applicability. Research on reliability suggests that CJ requires a relatively modest number of pairwise comparisons (Verhavert et al., 2019; Crompvoets et al., 2022) to generate trait scores that are as precise and consistent as those generated by other assessment methods (Coertjens et al., 2017; Goossens and De Maeyer, 2018; Bouwer et al., 2023). In addition, the evidence suggests that the reliability and time efficiency of CJ are comparable, if not superior, to those of other assessment methods when employing adaptive comparison algorithms (Pollitt, 2012b; Verhavert et al., 2022; Mikhailiuk et al., 2021). Meanwhile, research on validity indicates the capacity of CJ scores to represent accurately the traits under measurement (Whitehouse, 2012; van Daal et al., 2016; Lesterhuis, 2018; Bartholomew et al., 2018; Bouwer et al., 2023). Lastly, research on its practical applicability highlights CJ's versatility across educational and non-educational contexts (as seen in Maydeu-Olivares and Böckenholt, 2005; Kimbell, 2012; Jones and Inglis, 2015; Bartholomew et al., 2018; Jones et al., 2019; ZuccoJr. et al., 2019; Marshall et al., 2020; Bartholomew and Williams, 2020; Boonen et al., 2020; Pritikin, 2020; Schramm and Batchelder, 2021; Seymour et al., 2022; Seymour and Hernandez, 2025).

Nevertheless, despite the growing number of CJ studies, research in this area remains unsystematic and fragmented. Consequently, several critical issues remain unresolved. To address this gap, this study pursues two complementary aims. The first is to examine three prominent issues relevant to CJ applications in education, linguistics, and psychology. Specifically, it aims to discuss (1) the reliance on strong simplifying assumptions in the modeling of CJ data, (2) the limited attention given to relevant CJ assessment design features and their role in the modeling process, and (3) the separation of trait measurement from statistical inference. Collectively, these issues have the potential to compromise the reliability and validity of the resulting trait estimates (Zimmerman, 1994; Perron and Gillespie, 2015; McElreath, 2020), as well as the accuracy and precision of statistical inferences (McElreath, 2020; Kline, 2023; Hoyle, 2023).

Building on this examination, the study's second aim is to address the outlined issues by extending Thurstone's theory through a systematic and integrated approach that combines Causal and Bayesian inference methods. This extension integrates Thurstone's core theoretical principles with key design features commonly observed in contemporary CJ applications.

Beyond the potential for improving measurement reliability and validity and increasing the accuracy and precision of statistical inferences, this approach offers two additional advantages. First, it clarifies the interactions among all actors and processes involved in CJ assessments. Second, it shifts the comparative data analysis paradigm from passively accepting Case V and BTL model assumptions to actively testing whether these assumptions fit the observed data.

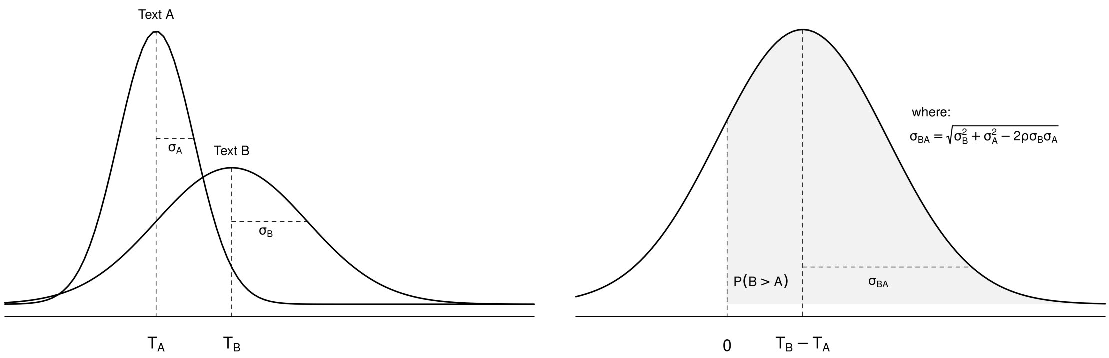
The remainder of this study is organized into seven sections. Section 2 provides an overview of Thurstone's theory. Section 3 examines the prominent issues identified from the CJ literature. Section 4 extends Thurstone's general form to address these issues. The extension integrates core theoretical principles alongside key CJ assessment design features. Section 5 translates these theoretical and design elements into a probabilistic and statistical model to analyze dichotomous pairwise comparison data. Section 6 reviews the findings, outline directions for future research, discusses the study's limitations and details the challenges that applied researchers may encounter. Finally, Section 7 provides the concluding remarks.

2. Thurstone's theory

In its general form, Thurstone's theory addresses pairwise comparisons of a single judge who evaluates multiple stimuli (Thurstone, 1927a,b). The theory posits that two key factors determine the dichotomous comparison outcome: the discriminal process of each stimulus and their discriminal difference. The *discriminal process* captures the psychological impact each stimulus exerts on the judge or, more simply, his perception of the stimulus trait. The theory assumes that the discriminal process for any given stimulus forms a Normal distribution along the trait continuum (Thurstone, 1927b). The mode (mean) of this distribution, known as the *modal discriminal process*, indicates the stimulus position on this continuum, while its dispersion, referred to as the *discriminal dispersion*, reflects variability in the perceived trait of the stimulus.

These ideas become clearer through an example. Figure 1a illustrates the hypothetical discriminal processes for two written texts on a *quality* trait continuum. The figure shows that the modal discriminal process for Text B lies further along the continuum than that of Text A ($T_B > T_A$), indicating that Text B has higher perceived quality. The figure also shows that Text B exhibits greater variability in its perceived quality than Text A, as reflected in its broader distribution driven by its larger discriminal dispersion ($\sigma_B > \sigma_A$).

However, because researchers cannot directly observe discriminal processes, Thurstone (1927a) introduced the *law of comparative judgment*. This law posits that, in pairwise comparisons, judges perceive the stimulus whose discriminal process lies further along the trait continuum as expressing



(a) Discriminal processes

(b) Discriminal difference

Figure 1: Hypothetical discriminal processes and discriminant difference along a quality trait continuum for two written texts.

more of the trait (Bramley, 2008). Consequently, comparison outcomes depend on the relative distance between stimuli rather than on their absolute positions on the trait continuum. Thurstone formalized this principle through the *discriminal difference*–defined as the difference between the discriminal processes of the stimuli under comparison–which ultimately determines the observed dichotomous outcome.

Notably, because the theory assumes that each discriminal process forms a Normal distribution on the trait continuum, the discriminal difference will also conform to a Normal distribution (Andrich, 1978). The mean (mode) of this distribution represents the average relative separation between the stimuli ($T_B - T_A$), while its dispersion $\sigma_{BA} = \sqrt{\sigma_B^2 + \sigma_A^2 - 2\rho\sigma_B\sigma_A}$ captures the variability of that separation, where ρ denotes the correlation between the two stimuli.

Figure 1b illustrates the distribution of the discriminal difference for the previous two hypothetical texts. The figure indicates that judges are more likely to perceive Text B as higher in quality than Text A for two reasons: the positive difference between their modal discriminal processes ($T_B - T_A > 0$) and the probability mass where the discriminal difference distinctly favors Text B over Text A, represented by the shaded gray area denoted as $P(B > A)$, which depends on the dispersion of the discriminal difference σ_{BA}. As a result, the dichotomous outcome of this comparison is more likely to favor Text B over A.

Notably, despite its conceptual appeal and formal elegance, Thurstone noted from the outset that this general formulation leads to a *trait scaling problem*. Specifically, the model requires estimating more "unknown" parameters than the number of available pairwise comparisons (Thurstone, 1927b). For example, in a CJ assessment with five texts, the general form requires estimating 20 parameters: five modal discriminal processes, five discriminal dispersions, and 10 correlations–one per comparison (see Table 1). However, a single judge provides only $\binom{5}{2} = 10$ unique comparisons, resulting in an

Table 1: Thurstones cases and their asumptions

Assumption	General form	Thurstone's Case I	Case II	Case III	Case IV	Case V	BTL model
Discriminal process (distribution)	Normal	Normal	Normal	Normal	Normal	Normal	Logistic
Discriminal dispersion (between stimuli)	Different	Different	Different	Different	Similar	Equal	Equal
Correlation (between stimuli)	One per pair	Constant	Constant	Zero	Zero	Zero	Zero
How many judges compare?	Single	Single	Multiple	Multiple	Multiple	Multiple	Multiple

insufficient number of observations to estimate the full parameter set.

To address this issue and facilitate the practical implementation of his theory, Thurstone derived five cases from the general form, each progressively incorporating additional simplifying assumptions (Thurstone, 1927b). In Case I, Thurstone postulated that pairs of stimuli would maintain a constant correlation across all comparisons. In Case II, he allowed multiple judges to undertake comparisons instead of confining evaluations to a single judge. In Case III, he assumed no correlation between stimuli. In Case IV, he posited that the stimuli exhibited similar dispersions. Finally, in Case V, he replaced this condition with the assumption that stimuli had equal discriminal dispersions. Table 1 summarizes the assumptions of the general form and the five cases. For a detailed discussion of these cases and their growing simplification, refer to Thurstone (1927b) and Bramley (2008).

3. Three prominent issues in traditional CJ practice

As Table 1 suggests, Thurstone developed Case V with an emphasis on statistical simplicity. However, this simplicity comes at the expense of leaving several critical issues unresolved. Three issues are particularly relevant to CJ applications in education, linguistics, and psychology: (1) the reliance on strong simplifying assumptions in the modeling of CJ data; (2) the limited attention given to relevant CJ assessment design features and their role in the modeling process; and (3) the separation of trait measurement from statistical inference. The following sections examine the implications of these issues.

3.1. The reliance on strong simplifying assumptions in the modeling of CJ data

Thurstone's Case V remains the most widely used model in the CJ literature, and specially in education (Kimbell, 2012; Jones and Inglis, 2015; van Daal et al., 2016; Bartholomew et al., 2018), linguistics (Boonen et al., 2020), and psychology (Maydeu-Olivares and Böckenholt, 2005). Researchers largely favor it because of the widespread adoption of the Bradley-Terry-Luce (BTL) model (Bradley and Terry, 1952; Luce, 1959), which provides a simplified statistical representation of the Case. The BTL model incorporates most of Case V's assumptions, with one notable exception. While Case V assumes that the discriminal processes of stimuli follow a Normal distribution, the BTL model assumes the more mathematically tractable Logistic distribution (Andrich, 1978; Bramley, 2008), as seen in Table 1.

This substitution, however, has minimal impact on trait estimation or model interpretation because the scale of the discriminal process or stimulus trait is arbitrary up to a non-monotonic transformation (van der Linden, 2017a; McElreath, 2021). In other words, as long as a transformation preserves the rank order of the data, the choice of distribution for the discriminal processes is inconsequential. The BTL model satisfies this condition because the Normal and Logistic distributions exhibit analogous statistical properties, differing only by a scaling factor of approximately 1.7 (van der Linden, 2017a).

However, reliance on the BTL model–and by extension on Case V–raises concerns about the reliability and validity of trait estimates in contexts where model assumptions may not hold (Kelly et al., 2022; Andrich, 1978). Indeed, Thurstone himself cautioned that researchers should not use Case V "without (an) experimental test" (Thurstone, 1927b, p. 270), further acknowledging that some of its assumptions may be problematic when assessing complex traits or heterogeneous stimuli (Thurstone, 1927a). Consequently, because these conditions are common in contemporary CJ applications, two key assumptions of Case V and the BTL model may not hold in theory or practice: the assumptions of equal discriminal dispersions and zero correlations between stimuli.

3.1.1. The assumption of equal dispersions between stimuli

According to the theory, differences in the discriminal dispersions of stimuli shape the distribution of the discriminal difference, thereby directly influencing the outcome of pairwise comparisons. A thought experiment helps illustrate this idea. Suppose researchers observe the discriminal processes for two texts, A and B, where the dispersion for Text A remains constant and the two texts are uncorrelated ($\rho = 0$). As Figure 2 shows, an increase in the uncertainty associated with the perception of Text B relative to Text A ($\sigma_B - \sigma_A$) broadens the distribution of their discriminal difference. This broadening affects the probability area where the discriminal difference distinctly favors Text B over A, expressed as $P(B > A)$, ultimately influencing the comparison outcome.

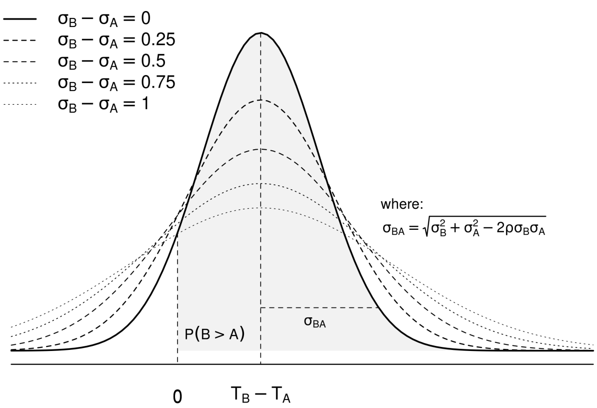


Figure 2: Discriminal Difference distribution under varying discrepancies in stimuli dispersions

In experimental practice, however, the thought experiment occurs in reverse. Researchers first observe the comparison outcome and then use the BTL model to infer the discriminal difference between stimuli and their respective discriminal processes (Thurstone, 1927a). Consequently, the effectiveness of the outcome to reflect *true* differences between stimuli largely depends on the validity of the model's assumptions (Kohler et al., 2019), in this case, the assumption of equal dispersions. When the assumption accurately captures the complexity of the data, the BTL model estimates a discriminal difference distribution that accurately represents the *true* discriminal difference between the texts. As illustrated in Figure 2, this occurs when the model-implied distribution aligns with the true discriminal difference distribution, represented by the thick continuous line corresponding to $\sigma_B - \sigma_A = 0$, and it is this alignment that supports reliable estimates of the underlying discriminal processes.

Notably, although the assumption of equal dispersions simplifies the trait measurement model, recurring discussions in the CJ literature suggest that it may not hold for complex trait or heterogeneous stimuli (Andrich, 1978; Bramley, 2008; Kelly et al., 2022). Indeed, Thurstone did not expect this assumption to hold except for the simplest stimuli. He noted that "the discriminal dispersion on the psychological continuum is of relatively less serious importance in dealing with strictly homogeneous stimulus series, but it becomes a serious factor in dealing with less conspicuous attributes or with less homogeneous stimulus series, such as handwriting specimens, English compositions, …" (Thurstone, 1927a, pp. 376).

However, more relevant to this discussion are the statistical and measurement issues that arise when the assumption of equal dispersions between stimuli does not hold. One such issue is the overestimation of trait reliability under the BTL model. This overestimation can lead to spurious conclusions about discriminal differences (McElreath, 2020; Wu et al., 2022) and, by extension, about the underlying discriminal processes. Reliability here refers to the extent to which observed outcomes reflect the true discriminal differences between stimuli. As Figure 2 illustrates, this overestimation occurs when the assumed discriminal difference (represented by the thick continuous line under the BTL assumption $\sigma_B - \sigma_A = 0$) does not match the *true* discriminal difference, which may follow any alternative distribution where $\sigma_B - \sigma_A \neq 0$.

A second issue concerns the over-identification and exclusion of misfit texts. *Misfit texts* refer to stimuli that elicit more judgment discrepancies than others (Pollitt, 2004, 2012b,a; Goossens and De Maeyer, 2018). If these discrepancies arise from comparisons involving stimuli with differing discriminal dispersions, driven by stimulus heterogeneity, then the common CJ practice of excluding stimuli based on such statistics (Pollitt, 2012a,b) may unintentionally discard valuable information (Miller, 2023) and introduce bias into trait estimates (Zimmerman, 1994; McElreath, 2020). The direction and magnitude of this bias remain unpredictable, as they depend on which stimuli are removed from the analysis.

3.1.2. The assumption of zero correlation between stimuli

According to the theory, the correlation between two stimuli (ρ) captures the extent to which judges' perception of a given trait in one stimulus depends on their perception of the same trait in the other stimulus (Thurstone, 1927a). Like discriminal dispersions, this correlation shapes the distribution of the discriminal difference, and thus, directly influences pairwise comparison outcomes.

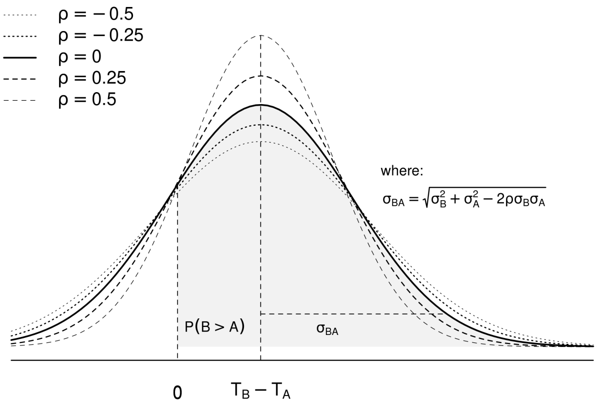


Figure 3: Discriminal Difference distribution under varying levels of correlation between stimuli

Building on the previous thought experiment, suppose researchers observe the discriminal processes for texts A and B, and are able to hold their discriminal dispersions constant. As Figure 3 shows, increasing correlation narrows the distribution of the discriminal difference, whereas decreasing correlation widens it. This narrowing or widening, in turn, changes the probability that the discriminal difference favors Text B over Text A, denoted as $P(B > A)$, and thus directly influences the comparison outcome.

In experimental practice, however, this process again proceeds in reverse: researchers use the BTL model to infer the shape of the discriminal difference from observed comparison outcomes. Consequently, the validity of the model's assumptions–in this case, the zero correlation assumption–becomes central to whether the inferred traits reflect the true underlying data-generating process (Kohler et al., 2019).

Notably, contemporary CJ assessment designs often adopt the assumption of zero correlation between stimuli based on an early theoretical argument proposed by Thurstone (1927b). This argument rests on the idea that the absence of systematic judge bias is sufficient for the assumption to hold. Specifically, Thurstone argued that stimuli could be treated as uncorrelated because judges' biases–arising from two opposing and equally weighted effects occurring during pairwise comparisons–would cancel each other out. Andrich (1978) later formalized this argument by mathematically demonstrating the cancellation of additive biases within the BTL model. However, evidence from the CJ literature indicates that the assumption of zero correlation does not hold in practice in at least two cases: (1) when intricate aspects of multidimensional, complex traits or heterogeneous stimuli influence judges' perceptions, (2) when additional hierarchical structures are relevant to the stimuli.

Regarding the first case, research on text quality assessments suggests that when judges evaluate complex, multidimensional traits or heterogeneous stimuli, they often rely on a variety of intricate stimulus characteristics to inform their judgments (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Regardless of their relevance, these characteristics may not be equally weighted or consistently opposed across comparisons. As a result, they may exert a disproportionate influence on judges' perceptions, generating biases that persist rather than cancel out. For example, this might occur when a judge assessing the argumentative quality of a text is disproportionately influenced by the clarity of the handwriting, thereby favoring neatly written texts even if their arguments are weaker.

Moreover, because researchers can only infer the discriminal process of stimuli through judges' perceptions–and these perceptions may carry some degree of bias or systematic deviation from consensus (see Pollitt and Elliott, 2003; van Daal et al., 2016)–such biases can introduce dependencies between stimuli (van der Linden, 2017b), directly violating the assumption of zero correlation. Although direct evidence for this specific mechanism remains limited, existing studies document the presence of judge biases in CJ contexts (Pollitt and Elliott, 2003; van Daal et al., 2016; Bartholomew et al., 2020), reinforcing the argument that the factors influencing pairwise comparisons do not always cancel out.

In the second case, the shared context or inherent connections introduced by additional hierarchical structures may create dependencies between stimuli–a statistical phenomenon known as clustering (Everitt and Skrondal, 2010). For instance, when the same individual produces multiple texts, those texts often share several features such as writing style or overall quality, that clearly violate the assumption of no correlation. Notably, although some CJ studies acknowledge the presence of such hierarchical structures and account for them (e.g., Pritikin, 2020; Schramm and Batchelder, 2021; Seymour et al., 2022; Seymour and Hernandez, 2025), the treatment of this additional source of dependence in other research has often been insufficient. For instance, when CJ data include multiple samples of stimuli from the same individuals, researchers frequently rely on BTL mean trait estimates to conduct follow-up analyses at the individual level (van Daal et al., 2017; Jones et al., 2019; Bramley and Vitello, 2019; Boonen et al., 2020; Gijsen et al., 2021; Bouwer et al., 2023), which introduces additional issues (see Section 3.3).

More importantly, erroneously assuming zero correlation between stimuli also leads to significant statistical and measurement issues. In particular, neglecting judges' biases or relevant hierarchical structures can create dimensional mismatches in the model, leading to the over- or under-estimation of the trait and its reliability (Reckase et al., 1986; Ackerman, 1989; Kahraman, 2013; Hoyle, 2023). These inaccuracies can produce spurious results and misleading conclusions about discriminal differences (McElreath, 2020) and, by extension, about the underlying discriminal processes of the stimuli.

One such spurious result may be the incorrect classification of stimuli as *misfits* due to the violation of the zero correlation assumption, which parallels the scenario described in the previous section. This misclassification arises because the BTL model cannot capture how judges perceive stimuli as similar or dissimilar during comparisons. As Figure 3 illustrates, this occurs when the discriminal difference distribution implied by the BTL model–represented by the thick continuous line $\rho = 0$–differs from the *true* discriminal difference, which may follow any alternative distribution where $\rho \neq 0$. As discussed in the previous section, removing these misfit stimuli risks discarding valuable information and introducing bias into trait estimates (Miller, 2023).

In any case, Case V and the BTL model have limited capacity to accommodate differences in discriminal dispersions or to account for correlations among stimuli. As a result, researchers may overestimate trait reliability, exclude "problematic" stimuli when they are identified as misfits, and draw spurious conclusions about the trait of interest due to the models' limited flexibility.

3.2. The lack of consideration of CJ assessment design features

Setting aside their limited capacity to accommodate more realistic assumptions, Case V and the BTL model also lack mechanisms to incorporate key design features commonly observed in contemporary CJ assessments. Specifically, their limited flexibility prevents them from directly assessing the presence of judge bias or from characterizing the role of sampling and comparison algorithms in trait estimation and statistical inference.

With respect to judges' bias, Section 3.1.2 shows that the no-bias assumption becomes implausible when judges evaluate complex, multidimensional traits or heterogeneous stimuli, a common scenario in contemporary CJ applications (see, van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022). Nevertheless, despite this implausibility, the CJ literature often relies on ad hoc procedures that use BTL-derived trait scores, their transformations, or residuals to evaluate such bias. For example, Wu (2025) fitted an analysis of variance (ANOVA) model to judges' infit statistics to examine whether judge expertise influences judgment behavior. These infit statistics–defined as weighted averages of squared Pearson residuals–indicate the extent to which judges' decisions deviate from model-based expectations relative to other judges (Wright and Masters, 1982). Similar approaches appear in Pollitt and Elliott (2003), van Daal et al. (2016), and Bartholomew et al. (2020).

However, these approaches are difficult to validate because they rely on circular reasoning: they attempt to assess judges' bias–an additional latent dimension (Hoyle, 2023)–using outputs from a model that assumes no such bias exists. Consequently, the validity of these methods depends on the absence of the very phenomenon they seek to evaluate.

Luckily, the psychometric literature already documents the consequences of this type of model misspecification. When a latent model omits substantively important dimensions of the data-generating process, it produces systematically misleading outputs. These misleading outputs occur because misspecified models redistribute variation from omitted dimensions into the parameters they do estimate (Reckase et al., 1986; Ackerman, 1989; Kahraman, 2013; Hoyle, 2023). In CJ assessments, this issue may arise, for example, when researchers use BTL residuals to evaluate judge bias, even though the model explicitly assumes that no such bias exist. As a result, the model may not separate bias from the latent trait structure and may instead absorb part of the bias into the trait estimates. This absorption distorts the trait and residual structure, causing residual-based analyses to overestimate or underestimate judge bias, misrepresent its direction, or fail to detect it altogether (Kelly et al., 2022).

Similarly, regarding the role of sampling and comparison algorithms in trait estimation and statistical inference, the CJ literature often downplays their importance by relying on a key, though often implicit, assumption of Case V and the BTL model: the "sample-free" or "population-free" calibration assumption. This assumption states that different samples drawn from the same empirical population, whether random or not, should produce the same relative scale estimates (Wright, 1977; Andrich, 1978). In other words, the estimated distance between two stimuli does not depend on the other stimuli with which they are compared (Bramley, 2008).

Notably, the literature already stated that "this property of the model is true in empirical data only if the model holds in the data" (Andrich, 1978, p. 459). Despite this caveat, the CJ literature has often taken the sample-free assumption as given. More importantly, it has extended this assumption beyond its original scope to the interpretation that researchers do not need appropriate paired-comparison designs, because the estimation procedure can accommodate even non-random missing data (Bramley, 2008). However, a substantial body of statistical literature shows that sampling and other mechanisms, that reduce observed data or induce missingness, can affect trait estimation and statistical inference (see, Kohler et al., 2019; McElreath, 2020, chap. 15; Schuessler and Selb, 2023).

In summary, ignoring additional latent dimensions when modeling CJ data–such as judges' bias–can yield biased trait estimates and residuals (Reckase et al., 1986; Ackerman, 1989; Kahraman, 2013; Hoyle, 2023). Similarly, assuming that all CJ assessment designs inherently satisfy the sample-free assumption not only obscures the role of sampling and comparison algorithms within the assessment system but can also affect trait estimation and statistical inference (Kohler et al., 2019; McElreath, 2020, chap. 15; Schuessler and Selb, 2023).

3.3. The disconnect between trait measurement and statistical inference

The third and final concern regarding the use of Case V and the BTL relates to the separation between trait measurement and statistical inference. Contemporary CJ applications across education, linguistics, and psychology show that follow-up analyses–which typically involve statistical inference–often rely on point estimates of BTL trait scores, their transformations, or derived residuals. Researchers use these outputs, for instance, to identify misfitting judges and stimuli (Pollitt, 2012b; van Daal et al., 2016; Goossens and De Maeyer, 2018), detect bias in judges' ratings (Pollitt and Elliott, 2003; Pollitt, 2012b; Wu, 2025), examine correlations with other assessment methods (Goossens and De Maeyer, 2018; Bouwer et al., 2023), or draw statistical inferences about the underlying trait (Casalicchio et al., 2015; Bramley and Vitello, 2019; Boonen et al., 2020; Bouwer et al., 2023; van Daal et al., 2017; Jones et al., 2019; Gijsen et al., 2021).

However, this practice reflects a historical gap in Thurstone's original framework, which provided no guidance for formal statistical inference. Specifically, this limitation arose because Thurstone's primary goal was to produce a "rather coarse scaling" of traits and to allocate compared stimuli along such a continuum (Thurstone, 1927b, p. 269), and little beyond that. As a result, the CJ tradition attempted to bridge this inference gap by separating trait measurement from statistical inference.

Nevertheless, the statistical literature cautions against separating these two processes. Specifically, while the separation simplifies the analysis of CJ data, it also prevents the propagation of uncertainty from estimation to inference (Pritikin, 2020), ultimately undermining the validity of the resulting conclusions. A key consideration is that BTL trait scores are parameter estimates that inherently carry uncertainty (measurement error). Ignoring this uncertainty can bias parameter estimates and reduce their precision. The direction and magnitude of such biases are often unpredictable. Results may be attenuated, exaggerated, or remain unaffected depending on the degree of uncertainty in the scores and the actual effects being tested (McElreath, 2020; Kline, 2023; Hoyle, 2023). Moreover, reduced precision in trait estimates diminishes statistical power, increasing the risk of both type-I and type-II errors in inferences (McElreath, 2020).

In summary, the heavy reliance on strong simplifying assumptions in the modeling of CJ data, together with limited attention to CJ assessment design features and the disconnect between trait measurement and statistical inference, can compromise the reliability and validity of trait estimates (Zimmerman, 1994; Perron and Gillespie, 2015; McElreath, 2020) and reduce the accuracy and precision of statistical inferences (McElreath, 2020; Kline, 2023; Hoyle, 2023).

The structural approach to Causal inference provides a systematic and integrated framework for extending Thurstone's theory beyond these limitations of Case V and the BTL model.

4. Extending Thurstone's theory

The *structural approach* to Causal inference (Pearl, 2009; Pearl et al., 2016) offers a formal framework for identifying causes and estimating their effects from data. Specifically, the approach represents the assumed causal structure of a system, such as that underlying CJ assessments, using *directed acyclic graphs (DAGs)* and *structural causal models (SCMs)* (Morgan and Winship, 2014; Gross et al., 2018; Neal, 2020). DAGs and SCMs function as conceptual models that support *identification* (Schuessler and Selb, 2023). Identification determines whether an estimator can accurately compute a target estimand based solely on causal assumptions, independent of random variation. Here, *estimands* represent the specific quantities researchers aim to determine (i.e., a parameter) (Everitt and Skrondal, 2010). *Estimators* denote the methods or functions that transform data into an estimate (e.g., a statistical model), while *estimates* are the numerical values approximating the estimand (Neal, 2020; Everitt and Skrondal, 2010).

To illustrate these concepts, consider a researcher who seeks to answer the question: "To what extent do different teaching methods influence students' ability to produce high-quality written texts?' To investigate this question, the researcher designs a CJ assessment and randomly assigns students (the individuals) to two groups, each receiving a different teaching method. Judges then compare pairs of students' written texts (the stimuli) and produce a dichotomous outcome that reflects the relative quality of each text (the trait). Based on this design, researchers can reformulate the research question as the estimand: *On average, is there a difference in the ability to produce high-quality written texts between the two groups of students?* Following common practice in education, linguistics, and psychology, researchers typically use BTL mean trait estimates or their transformations to approximate this estimand (see Section 3.3).

However, as discussed in the previous section, Thurstone's Case V and the BTL model exhibit several statistical and measurement limitations when applied to contemporary CJ assessments. These limitations can hinder the model's ability to identify and accurately estimate a range of estimands relevant to CJ inquiries, including the one described in the previous illustration.

Fortunately, DAGs and SCMs support identification through two key advantages[1]. First, they represent causal structures of any complexity using a small set of fundamental building blocks (Neal, 2020; McElreath, 2026). This feature enables the decomposition of complex systems into simpler components that facilitate analysis. Second, they encode causal relationships in a non-parametric form. This flexibility enables feasible identification strategies without requiring specification of the

[1]In depth explanation of these topics is beyond the scope of this study, thus, readers seeking a more profound understanding can refer to introductory papers such as Pearl (2010), Rohrer (2018), Pearl (2019), and Cinelli et al. (2020), and introductory books like Pearl and Mackenzie (2018), Neal (2020), and McElreath (2020) are useful. For more advanced study, seminal papers such as Neyman (1923), Rubin (1974), Spirtes et al. (1991), and Sekhon (2009), along with books such as Pearl (2009), Morgan and Winship (2014), and Hernán and Robins (2025), are recommended.

types of variables, the functional forms relating them, or the parameters of those functional forms (Pearl et al., 2016)

Because these advantages remain abstract at this stage, the following sections illustrate how DAGs and SCMs extend Thurstone's theory beyond the limitations of Case V and the BTL model, while supporting identification. Specifically, Section 4.1 introduces the *conceptual-population model (CPM)*, which integrates Thurstone's theoretical principles into the DAG and SCM representation of the CJ system. The CPM assumes an idealized setting in which researchers observe the *conceptual population* of comparative judgment data. Moreover, Section 4.2 presents the *sample-comparison model (CSM)*, which incorporates key design features commonly observed in contemporary CJ assessments into the same DAG and SCM representations. The CSM assumes a more realistic setting in which researchers observe only a sample drawn from the conceptual population.

4.1. The conceptual-population model

The CPM assumes an idealized scenario in which researchers observe a *conceptual population* of comparative judgment data. This conceptual population includes all judgments made by every available judge for each pair of stimuli produced by each pair of individuals in the population. This setting provides a natural framework for integrating Thurstone's theoretical principles and for proposing innovations that address some of the issues discussed in Section 3.

4.1.1. Integrating the first theoretical principles

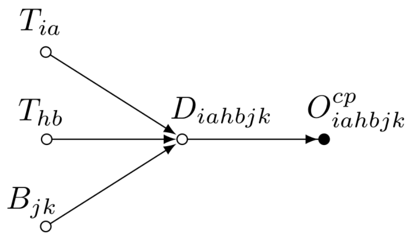
Building on the teaching-method example introduced at the beginning of Section 4, this section uses SCMs and DAGs to incorporate Thurstone's first theoretical principles into the CPM.

SCMs are formal mathematical models defined by a set of *endogenous* variables V, a set of *exogenous* variables E, and a set of functions F (Pearl, 2009; Pearl et al., 2016; Cinelli et al., 2020). Endogenous variables are those whose causal mechanisms a researcher chooses to model (Neal, 2020), while exogenous variables capture *errors* or *disturbances* arising from omitted factors that researchers choose not to explicitly model (Pearl, 2009; Pearl et al., 2016). The functions, referred to as *structural equations*, express each endogenous variables as non-parametric function of other endogenous and exogenous variables. These equations use the symbol '$:=$' to denote the asymmetrical causal relationship between variables, whereas the symbol '$\perp$' to denote *d-separation*, a concept akin to (conditional) independence (see Section 8.1.3.1).

Considering the teaching-method example, the SCM shown in Figure 4a describes the relationship between the conceptual-population outcome O^{cp}_{iahbjk}, the discriminal difference between the compared texts D_{iahbjk}, the texts' discriminal processes T_{ia} and T_{hb}, and the judgment-level biases B_{kj}, which represent systematic deviations of an individual judgment from the judge-specific expectations. Specifically, the comparison outcome is assumed to be causally influenced only by the discriminal

$$O^{cp}_{iahbjk} := f_O(D_{iahbjk})$$
$$D_{iahbjk} := f_D(T_{ia}, T_{hb}, B_{jk})$$

(a) SCM



(b) DAG

Figure 4: Conceptual-population model (CPM), scalar form.

difference, which itself is causally determined by the discriminal processes and judgment-level biases. Notably, the SCM expresses these relationships in non-parametric form through the structural equations f_O and f_D, meaning that it does not emphasize the types of variables, the functional forms relating them, or the parameters of those functional forms (Pearl et al., 2016). In this context, the endogenous variables set $V = \{O_{iahbjk}, D_{iahbjk}, T_{ia}, T_{hb}, B_{kj}\}$, while the structural equations set $F = \{f_O, f_D\}$.

Moreover, the subscripts i and h identify the students who authored the texts (i.e., the individuals). The indices a and b identify the texts under comparison (i.e., the stimuli). The index j represents the judge conducting the comparison, whereas k indexes judgments of the same stimulus pair by the same judge, as may occur in *repeated-measures designs* (Lawson, 2015). Thus, the indexing system supports comparisons between different texts written by the same student ($i = h$; $a \neq b$) and between texts written by different students ($i \neq h$; where $a = b$ is permitted), each compared once or repeatedly by more than one judge ($j > 1$; $k \geq 1$). However, it excludes cases where a judge compares a student's text to itself, whether once or multiple times ($i = h$; $a = b$; $j > 1$; $k \geq 1$), as such comparison lacks practical relevance within CJ assessments.

Notably, every SCM has an associated DAG (Pearl et al., 2016; Cinelli et al., 2020). A DAG is a *graph* consisting of nodes connected by edges, where nodes represent random variables. The term *directed* indicates that edges or arrows extend from one node to another, indicating the direction of causal influence. The absence of an edge implies no direct relationship between the nodes. The term *acyclic* means that the causal influences do not form loops, ensuring the influences do not cycle back on themselves (McElreath, 2020). By convention, DAGs depict observed variables as solid black circles and latent variables as open circles (Morgan and Winship, 2014). They may also include exogenous variables, as they often improve clarity and help reveal potential issues related to conditioning and confounding (Cinelli et al., 2020).

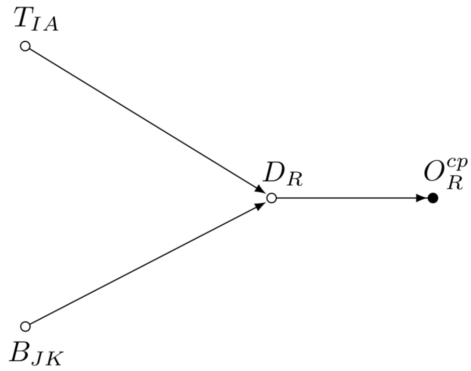
Figure 4b presents the DAG corresponding to the SCM in Figure 4a. The DAG depicts the causal relationships implied by Thurstone's theory: the texts' discriminal processes influence their discriminal difference, which in turn determines the pairwise comparison outcome. It also represents the influence of judgment-level biases on the discriminal difference. Finally, following convention, the DAG distinguishes observed variables, shown as solid black circles (e.g., the comparison outcome), from latent variables, shown as open circles (e.g., the texts' discriminal processes, the discriminal difference, and judgment-level biases).

4.1.2. The conceptual-population *data structure*

Although specifying a data structure is not mandatory when using SCMs and DAGs, we introduce one here to re-express the scalar form of the CJ system shown in Figure 4 as an equivalent vectorized representation, which improves clarity and facilitates the description of the system.

$$I = \begin{bmatrix} 1 \\ \vdots \\ i \\ \vdots \\ h \\ \vdots \\ n_I \end{bmatrix}; J = \begin{bmatrix} 1 \\ \vdots \\ j \\ \vdots \\ n_J \end{bmatrix}; IA = \begin{bmatrix} 1 & 1 \\ \vdots & \vdots \\ 1 & n_A \\ \vdots & \vdots \\ i & a \\ \vdots & \vdots \\ h & b \\ \vdots & \vdots \\ n_I & 1 \\ \vdots & \vdots \\ n_I & n_A \end{bmatrix}; JK = \begin{bmatrix} 1 & 1 \\ \vdots & \vdots \\ 1 & n_K \\ \vdots & \vdots \\ j & k \\ \vdots & \vdots \\ n_J & 1 \\ \vdots & \vdots \\ n_J & n_K \end{bmatrix}; R = \begin{bmatrix} 1 & 1 & 1 & 2 & 1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 2 & 1 & n_K \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ i & a & h & b & j & k \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ n_I & n_A - 1 & n_I & n_A & n_J & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ n_I & n_A - 1 & n_I & n_A & n_J & n_K \end{bmatrix} \tag{1}$$

For this purpose, we define the vectors I and J, along with the matrices IA and JK, as in Equation 1. Each element of I represents a unique individual $i = 1, \ldots, h, \ldots, n_I$, where n_I denotes the total number of individuals. Similarly, each element of J corresponds to a unique judge $j = 1, \ldots, n_J$, where n_J denotes the total number of judges. Moreover, matrix IA contains $n_I \cdot n_A$ rows and 2 columns, where n_A denotes the number of stimuli produced by an individual, and each row represents a unique individual-stimuli pairing (i, a) or (h, b). Likewise, matrix JK has $n_J \cdot n_K$ rows and 2 columns, where n_K denotes the number of judgments of the same stimulus pair per judge, and each row of the matrix represents judge-judgments pairings (j, k). Finally, we construct the matrix R to map each row of the IA matrix with a corresponding row from the JK matrix. This matrix has n rows and 6 columns, with $n = \binom{n_I \cdot n_A}{2} \cdot n_J \cdot n_K$. Here, the binomial coefficient $\binom{n_I \cdot n_A}{2}$ represents the total number of unique comparisons between all pairs of stimuli generated by all pairs of individuals in the population.



(b) DAG

$$O_R^{cp} := f_O(D_R)$$
$$D_R := f_D(T_{IA}, B_{JK})$$

(a) SCM

Figure 5: Conceptual-population model (CPM), initial vectorized form.

$$I = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} ; \; J = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} ; \; IA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \\ 2 & 2 \\ 3 & 1 \\ 3 & 2 \\ 4 & 1 \\ 4 & 2 \\ 5 & 1 \\ 5 & 2 \end{bmatrix} ; \; JK = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 2 & 1 \\ 2 & 2 \\ 2 & 3 \\ 3 & 1 \\ 3 & 2 \\ 3 & 3 \end{bmatrix} ; \; R = \begin{bmatrix} 1 & 1 & 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 1 & 3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 5 & 2 & 1 & 1 \\ 1 & 1 & 5 & 2 & 1 & 2 \\ 1 & 1 & 5 & 2 & 1 & 3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 4 & 2 & 5 & 2 & 3 & 1 \\ 4 & 2 & 5 & 2 & 3 & 2 \\ 4 & 2 & 5 & 2 & 3 & 3 \\ 5 & 1 & 5 & 2 & 3 & 1 \\ 5 & 1 & 5 & 2 & 3 & 2 \\ 5 & 1 & 5 & 2 & 3 & 3 \end{bmatrix} \tag{2}$$

With this data structure, the CJ system corresponding to the teaching-method example becomes easier to visualize. Figure 5 shows the equivalent vectorized form of the SCM and DAG depicted in Figure 4. In this representation, the outcome O_R^{cp}, the texts' discriminal difference D_R, their discriminal processes T_{IA}, and the judgment-level biases B_{JK} are represented as vectors rather than scalar values, capturing all observations from the conceptual population. Specifically, O_R^{cp} and D_R are observed and latent vectors of length n, respectively, while T_{IA} and B_{JK} are latent vectors of lengths $n_I \cdot n_A$ and $n_J \cdot n_K$, respectively. For illustration, Equation 2 shows how the matrices in

Equation 1 look under the assumption $n_I = 5$, $n_J = 3$, $n_A = 2$, and $n_K = 3$.

4.1.3. Integrating hierarchical structural components

Building on the principles of Structural Equation Modeling (SEM) (Hoyle, 2023) and Item Response Theory (IRT) (Fox, 2010; van der Linden, 2017a), the CPM integrates two *hierarchical structural components* to examine how *relevant observed and latent variables*[2] influence the primary latent variable of interest (Everitt and Skrondal, 2010). This hierarchical structure supports statistical inference that accounts for both the nested structure of stimuli and the uncertainty inherent in trait estimation (see Section 3.1.2 and Section 3.3 for a discussion of these considerations).

Figure 6b illustrates how the CPM integrates these two hierarchical components. Considering the endogenous variables set $V = \{O_R^{cp}, D_R, T_{IA}, B_{JK}, T_I, B_J, X_I, X_{IA}, Z_J, Z_{JK}\}$, the exogenous variables set $E = \{e_I, e_J, e_{IA}, e_{JK}\}$, and the structural equations set $F = \{f_O, f_D, f_T, f_B\}$, the top-left branch represents the first component, whereas the bottom-left branch represents the second.

In the top left branch, *relevant* student-level variables X_I (e.g., teaching method) and students' idiosyncratic errors e_I are assumed to causally influence the latent variable representing students' writing-quality ability T_I, while the error term e_I captures variation in students' traits that remains unexplained by X_I. In addition, this branch assumes that T_I, together with *relevant* text-level variables X_{IA} (e.g., text length) and texts' idiosyncratic errors e_{IA}, causally influence the texts' written-quality trait T_{IA}, which represents the first primary latent variable of interest. As before, the error term e_{IA} captures variation in text traits that remains unexplained by T_I or X_{IA}.

To clarify the structure of this branch, it is important to mention that X_I is an observed matrix with n_I rows and q_I independent columns, while both e_I and T_I are latent vectors of length n_I. Likewise, X_{IA} is an observed matrix with $n_I \cdot n_A$ rows and q_{IA} independent columns, while e_{IA} and T_{IA} are latent matrices with n_I rows and n_A columns.

Similarly, in the bottom left branch, *relevant* judge-level variables Z_J (e.g., judgment expertise) and judges' idiosyncratic errors e_J are assumed to causally influence the latent variable representing judges' bias B_J, while the error term e_J captures variation in judges' bias that remains unexplained by Z_J. Furthermore, this branch shows how B_J, together with *relevant* judgment-level variables Z_{JK} (e.g., the number of judgments a judge makes) and judgments' idiosyncratic errors e_{JK}, causally influence the judgment-level biases B_{JK}, which represents the second primary latent variable of interest. As before, the error term e_{JK} captures variation in judgments that remains unexplained by B_J or Z_{JK}.

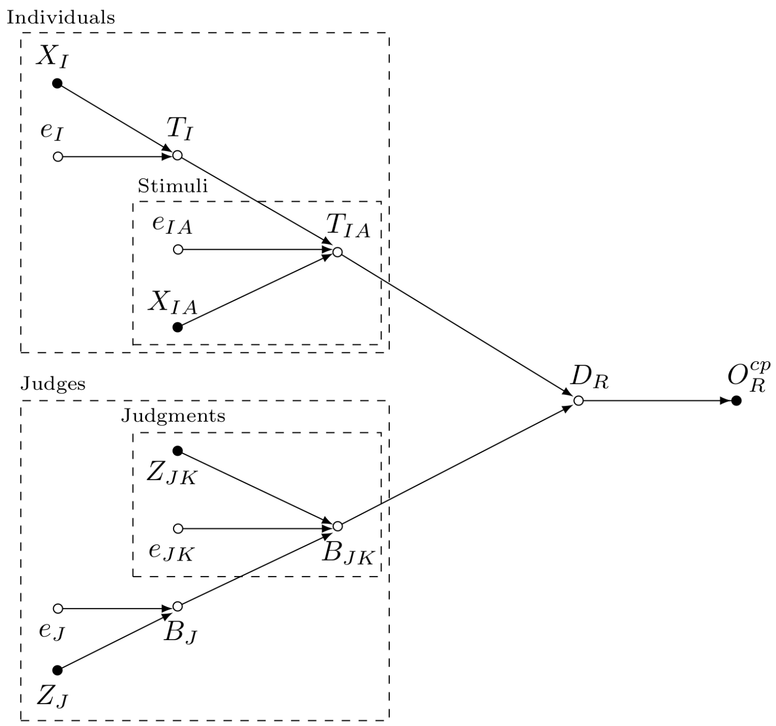
[2] *Relevant variables* are those that satisfy the *backdoor criterion* (Neal, 2020, pp 37), that is, they belong to a *sufficient adjustment set* (Pearl, 2009; Pearl et al., 2016; Morgan and Winship, 2014). A *sufficient* set (potentially empty) blocks all non-causal paths between a predictor and an outcome without opening new ones (Pearl, 2009). Refer also to footnote 1.

Here, Z_J is an observed matrix with n_J rows and q_J independent columns, while e_J and B_J are latent vectors of length n_J. Moreover, Z_{JK} is an observed matrix with $n_J \cdot n_K$ rows and q_{JK} independent columns, while e_{JK} and B_{JK} are latent matrices with n_J rows and n_K columns.

Finally, Figure 6a shows that all exogenous variables are assumed to be mutually independent, as indicated by the relationships $e_{IA} \perp \{e_I, e_{JK}, e_J\}$, $e_I \perp \{e_{JK}, e_J\}$, and $e_{JK} \perp e_J$, as well as by the absence of connecting arrows in Figure 6b.

$$
\begin{aligned}
O_R^{cp} &:= f_O(D_R) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 6: Conceptual-population model (CPM), final vectorized form.

Overall, the CPM extends Thurstone's theory by introducing two key innovations to address the limitations discussed in Section 3.1.2 and Section 3.3. These innovations include the explicit modeling of judge bias and the integration of hierarchical structural components. Nevertheless, despite its potential to improve measurement accuracy and precision, the CPM still assumes that researchers have access to data from the *conceptual population*. Since this assumption rarely holds in practice, we must consider a more realistic sampling scenario.

4.2. The sample-comparison model (CSM)

The CSM represents a more realistic scenario than the CPM. Specifically, Section 4.2.1 assumes that researchers observe only a sample of judgments ($n^s * K$) made by a sample of judges ($n^s * J$) on a subset of stimuli ($n^s * A$) produced by a sample of individuals ($n^s * I$), all drawn from the conceptual population. Furthermore, Section 4.2.2 assumes that judges do not perform every possible judgments within this sample. Instead, they complete only a sufficient number of pairwise comparisons, n_C, to allow accurate estimation of the proportion $P(B > A)$, as originally proposed by Thurstone (1927b).

4.2.1. The sample mechanism

To incorporate the *sampling mechanism* and facilitate the interpretation of the CSM, we first define the *data sampling process* using the binary vector variables S_I, S_J, S_{IA}, and S_{JK} as follows:

$$S_I = \begin{bmatrix} i_{(1)} \\ \vdots \\ i_{(i)} \\ \vdots \\ i_{(h)} \\ \vdots \\ i_{(nI)} \end{bmatrix} ; \; S_J = \begin{bmatrix} j_{(1)} \\ \vdots \\ j_{(j)} \\ \vdots \\ j_{(nJ)} \end{bmatrix} ; \; S_{IA} = \begin{bmatrix} ia_{(1,1)} \\ \vdots \\ ia_{(1,n_A)} \\ \vdots \\ ia_{(i,a)} \\ \vdots \\ ia_{(h,b)} \\ \vdots \\ ia_{(nI,1)} \\ \vdots \\ ia_{(nI,nA)} \end{bmatrix} ; \; S_{JK} = \begin{bmatrix} jk_{(1,1)} \\ \vdots \\ jk_{(1,n_K)} \\ \vdots \\ jk_{(j,k)} \\ \vdots \\ jk_{(nJ,1)} \\ \vdots \\ jk_{(nJ,nK)} \end{bmatrix} \tag{3}$$

Where each element of S_I is a binary value indicating the presence or absence of corresponding elements in vector I, as defined in Equation 4. We apply the same logic to the vector J to construct S_J (not shown). Accordingly, S_I and S_J contains n_I and n_J elements, respectively.

$$i_{(i)} = \begin{cases} 1 & \text{if data element } i \text{ from } I \text{ is sampled} \\ 0 & \text{if data element } i \text{ from } I \text{ is missing} \end{cases} \tag{4}$$

Similarly, each element of S_{IA} is a binary value indicating the presence or absence of data rows in the matrices IA, as defined in Equation 5. We apply the same logic to the matrix JK to construct

S_{JK} (not shown). Thus, S_{IA} and S_{JK} contains $n_I \cdot n_A$ and $n_J \cdot n_K$ elements, respectively.

$$ia_{(i,a)} = \begin{cases} 1 & \text{if data elements } i, a \text{ from } IA \text{ are sampled} \\ 0 & \text{if data elements } i, a \text{ from } IA \text{ are missing} \end{cases} \tag{5}$$

We illustrate the structure of these vectors with an example. Suppose researchers exclude the second student, the second text from each student, and the third judge from the conceptual population depicted in Equation 2. Given $n_I = 5$, $n_J = 3$, $n_A = 2$, and $n_K = 3$, the resulting vectors take the following form:

$$S_I = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} ; \; S_J = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} ; \; S_{IA} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} ; \; S_{JK} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{6}$$

Notably, Equation 6 shows that missing observations in vectors S_I and S_J–which represent unsampled students and judges–directly determine which observations are missing in S_{IA} and S_{JK}. In other words, researchers can only observe texts and judgments from students and judges initially included in the sample. The equation also shows that the sum of element in S_I equals the number of sampled students (n^s_I) and that a similar sum in S_J equals the sampled judges (n^s_J). Conversely, the sum of elements in S_{IA} represents the total sampled texts across all sampled students ($n^s_I \cdot n^s_A$), while a similar sum in S_{JK} represents the total sampled judgments across all sampled judges ($n^s_J \cdot n^s_K$).

Next, with the binary vector variables defined in Equation 3, we define the *sample mechanism* S as in Equation 7. This vector maps each element of S_{IA} to every element of S_{JK}, where each $s_{(i,a,h,b,j,k)}$ in S is a binary value indicating the presence or absence of data rows in the matrix R resulting from the sample mechanism depicted in Equation 8. Accordingly, the vector contains n elements, matching the number of rows in R, and its element sum represents the total data sample: $n^s = \binom{n^s_I \cdot n^s_A}{2} \cdot n^s_J \cdot n^s_K$. Here, $\binom{n^s_I \cdot n^s_A}{2}$ represents the binomial coefficient, which quantifies the total number of unique comparisons possible between every pair of sampled stimuli generated by each pair of sampled individuals.

$$S = \begin{bmatrix} s_{(1,1,1,2,1,1)} \\ \vdots \\ s_{(1,1,1,2,1,n_K)} \\ \vdots \\ s_{(i,a,h,b,j,k)} \\ \vdots \\ s_{(n_I,n_A-1,n_I,n_A,n_J,1)} \\ \vdots \\ s_{(n_I,n_A-1,n_I,n_A,n_J,1)} \end{bmatrix} \tag{7}$$

$$s_{(i,a,h,b,j,k)} = \begin{cases} 1 & \text{if data elements } i,a,h,b,j,k \text{ from } R \text{ are sampled} \\ 0 & \text{if data elements } h,i,a,b,j,k \text{ from } R \text{ are missing} \end{cases} \tag{8}$$

Following Bareinboim et al. (2014) and Schuessler and Selb (2023), Figure 7b incorporates the sampling mechanism into the CPM. The DAG depicts the conceptual-population outcome O_R^{cp} as an unobserved variable, highlighting its inaccessibility under the sampling mechanism. Moreover, it shows the *sample design* vector S as independent of all other variables–indicated by the absence of connecting arrows. This independence implies that Figure 7b applies exclusively to *Simple Random Sampling (SRSg)* designs, in which each judgment, judge, stimulus, and individual has an equal probability of inclusion within their respective group (Lawson, 2015). Finally, the square surrounding S denotes *conditioning*, that is, restricting the analysis to elements of the endogenous variable set V satisfying $s_{(i,a,h,b,j,k)} = 1$ (Neal, 2020; McElreath, 2020). In other words, S selects *all judgments made by a subset of judges for a subset of stimuli produced by the sampled individuals, all sampled under SRSg designs.*

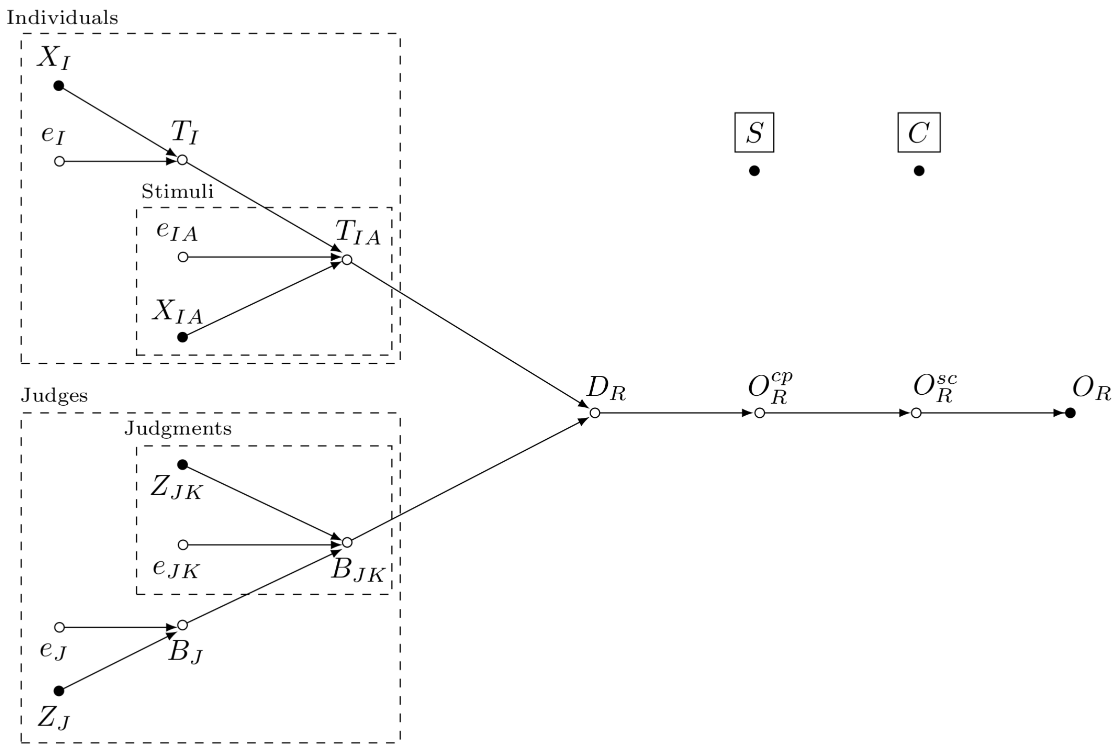
Nevertheless, due to concerns about the practical feasibility of the comparison task (Boonen et al., 2020), CJ assessments rarely implement an exhaustive pairings of sampled judges, stimuli, and individuals, as indicated above. Thus, a realistic scenario must account for the fact that judges typically compare only a subset of stimuli authored by a sample of individuals.

4.2.2. The comparison mechanism

As in the previous section, to incorporate the *comparison mechanism* and facilitate the interpretation of the CSM, we first define the *comparison process* using the binary vector variable C. Equation 9 shows that C contains n elements corresponding to the number of rows in the R matrix, with each element $c_{(i,a,h,b,j,k)}$ being a binary value indicating the presence or absence of data rows in R, a

$$
\begin{aligned}
O_R &:= f_C(O_R^{sc}, C)\\
O_R^{sc} &:= f_S(O_R^{cp}, S)\\
O_R^{cp} &:= f_O(D_R)\\
D_R &:= f_D(T_{IA}, B_{JK})\\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA})\\
T_I &:= f_T(X_I, e_I)\\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK})\\
B_J &:= f_B(Z_J, e_J)\\
e_I &\perp \{e_J, e_{IA}, e_{JK}\}\\
e_J &\perp \{e_{IA}, e_{JK}\}\\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 7: Sample-comparison model (CSM), vectorized form.

definition similar to that of $s_{(i,a,h,b,j,k)}$ in Equation 8.

$$C = \begin{bmatrix} c_{(1,1,1,2,1,1)} \\ \vdots \\ c_{(1,1,1,2,1,n_K)} \\ \vdots \\ c_{(i,a,h,b,j,k)} \\ \vdots \\ c_{(n_I,n_A-1,n_I,n_A,n_J,1)} \\ \vdots \\ c_{(n_I,n_A-1,n_I,n_A,n_J,1)} \end{bmatrix} \tag{9}$$

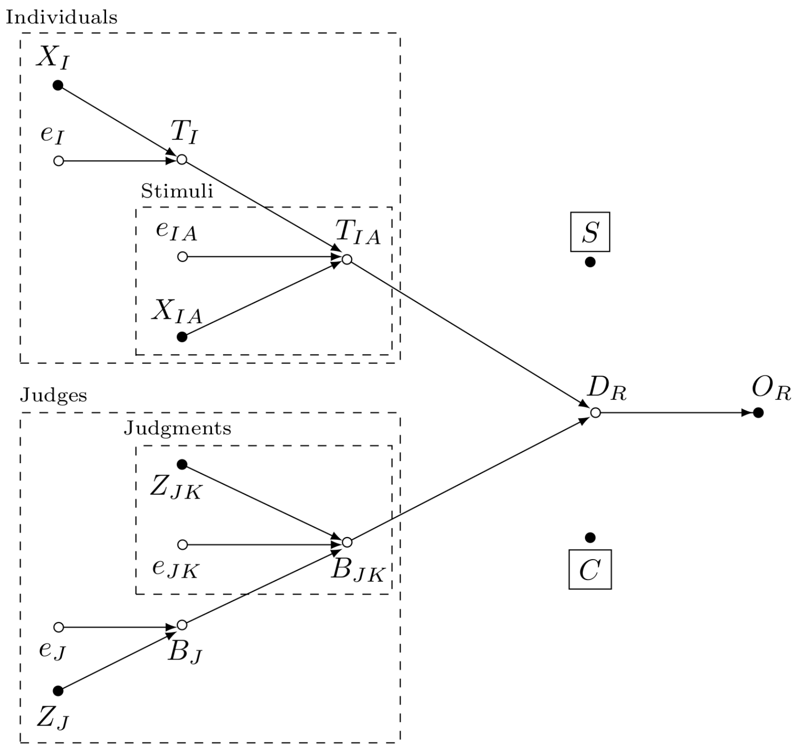
Again, following Bareinboim et al. (2014) and Schuessler and Selb (2023), Figure 7b incorporates the comparison mechanism into the CPM. The DAG represents the sample-comparison outcome O_R^{sc} as an unobserved variable, highlighting its inaccessibility under the comparison mechanism. Moreover, the *comparison design* vector C is shown as independent of all other variables, as indicated by the absence of connecting arrows. This independence implies that Figure 7b applies exclusively to *random allocation comparative design* (Bramley, 2015) or *incomplete block design* (Lawson, 2015), in which each pairwise comparison has an equal probability of being assigned to any judge. Finally, the square surrounding C denotes *conditioning*, that is, restricting the analysis to elements of the endogenous variable set V that also satisfy $c_{(i,a,h,b,j,k)} = 1$ (Neal, 2020; McElreath, 2020). In other words, C encodes the assumption that judges do not perform all possible judgments but instead complete a sufficient number, n_C, to enable accurate estimation of the proportion $P(B > A)$ for each stimulus pair (Thurstone, 1927b, p. 267).

Finally, since it is standard to assume that the distribution of the conceptual-population outcome O_R^{cp} also holds for O_R^{sc} and O_R, we reformulate the SCM and DAG in Figure 7 into the equivalent representations shown in Figure 8. In this latter version, we omit the unobserved outcomes O_R^{cp} and O_R^{sc}, and we assign the structural equation f_O–originally defined for O_R^{cp}–to O_R. In essence, this reformulation yields a model that applies directly to sampled CJ data, in which O_R depends directly on the discriminal difference D_R and is indirectly influenced by the sample and comparison mechanisms S and C (see Section 5).

In summary, the SCM and DAG in Figure 8 extends Thurstone's theory with innovations that address key limitations of Case V and the BTL model. It accounts for judgment- and judge-level biases, it reflects the hierarchical (clustered) structure of stimuli, thereby propagating the inherent uncertainty from trait estimation to inference, and clarifies the role of sample and comparison mechanisms in CJ assessments. However, despite these advances, the CSM does not resolve concerns about the assumption of equal dispersions among stimuli discussed in Section 3.1.1. Since this issue concerns the statistical assumptions underlying the distribution of the discriminal process, we develop a formal statistical model to address it in the next section.

$$
\begin{aligned}
O_R &:= f_O(D_R, S, C) \\
D_R &:= f_D(T_{IA}, B_{JK}) \\
T_{IA} &:= f_T(T_I, X_{IA}, e_{IA}) \\
T_I &:= f_T(X_I, e_I) \\
B_{JK} &:= f_B(B_J, Z_{JK}, e_{JK}) \\
B_J &:= f_B(Z_J, e_J) \\
e_I &\perp \{e_J, e_{IA}, e_{JK}\} \\
e_J &\perp \{e_{IA}, e_{JK}\} \\
e_{IA} &\perp e_{JK}
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 8: Comparative judgment model, vectorized form.

5. From SCM to statistical model

In this section we derive a statistical model that addresses violations of the equal dispersion assumption. To do so, we leverage the properties of SCMs to first factorize the joint distribution of our complex CJ system into a product of simpler *conditional probability distributions (CPDs)*[3]. Equation 10 shows the line-by-line equivalence between the SCM assumptions from Figure 8a and the resulting CPD factorization of the joint distribution. For clarity, we denote the joint distribution of variables L and M as $P(L, M)$, while we represent their conditional relationships using $P(L \mid M)$; $L \sim f(L \mid M)$; or $L := f_L(M)$.

$$
\begin{aligned}
P(O_R, &S, C, D_R, T_{IA}, X_{IA}, e_{IA}, T_I, X_I, e_I, B_{JK}, Z_{JK}, e_{JK}, B_J, Z_J, e_J) \\
&= P(O_R \mid D_R, S, C) \cdot P(S) \cdot P(C) \\
&\quad \cdot P(D_R \mid T_{IA}, B_{JK}) \\
&\quad \cdot P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \cdot P(X_{IA}) \\
&\quad \cdot P(T_I \mid X_I, e_I) \cdot P(X_I) \\
&\quad \cdot P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \cdot P(Z_{JK}) \\
&\quad \cdot P(B_J \mid Z_J, e_J) \cdot P(Z_J) \\
&\quad \cdot P(e_{IA}) \cdot P(e_I) \cdot P(e_{JK}) \cdot P(e_J)
\end{aligned} \tag{10}
$$

Notably, we can further simplify the factorization in Equation 10 into Equation 11 by making two observations. First, the mutually independent variables S, C, X_{IA}, X_I, Z_{JK}, and Z_J are observed data, implying that their probability distributions $P(S)$, $P(C)$, $P(X_{IA})$, $P(X_I)$, $P(Z_{JK})$, and $P(Z_J)$ require no further distributional modeling–in the same way as the idiosyncratic error terms do–since they contribute only constant factors to the overall joint distribution. Second, owing to the underlying random selection procedures[4]–namely, simple random sampling (SRSg) and random allocation comparative designs–the CPD of the outcome can be re-expressed as $P(O_R \mid D_R, S, C) = P(O_R \mid D_R)$.

[3]This re-expression is possible because the *chain rule* of probability and the *Bayesian Network Factorization (BNF)* property. For further details, see Pearl et al. (2016) and Neal (2020).

[4]Randomization ensures that data–and, by extension, an estimator–satisfies several key identification properties, such as common support, no interference, and consistency. The most critical property, however, is the elimination of confounding. *Confounding* occurs when an external variable, such as X_I, simultaneously influences both the outcome (e.g., O_R) and a variable of interest (e.g., S), resulting in spurious associations between the latter two (Everitt and Skrondal, 2010). Randomization ensure the absence of confounding by effectively decoupling the association between the variable of interest and any other variable, except for the outcome itself. For a more detailed discussion on the benefits of randomization, see Pearl (2009), Morgan and Winship (2014), Neal (2020), and Hernán and Robins (2025).

$$
\begin{aligned}
P(O_R,&S, C, D_R, T_{IA}, X_{IA}, e_{IA}, T_I, X_I, e_I, B_{JK}, Z_{JK}, e_{JK}, B_J, Z_J, e_J) \\
&= P(O_R \mid D_R) \\
&\quad \cdot P(D_R \mid T_{IA}, B_{JK}) \\
&\quad \cdot P(T_{IA} \mid T_I, X_{IA}, e_{IA}) \\
&\quad \cdot P(T_I \mid X_I, e_I) \\
&\quad \cdot P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) \\
&\quad \cdot P(B_J \mid Z_J, e_J) \\
&\quad \cdot P(e_{IA}) \cdot P(e_I) \cdot P(e_{JK}) \cdot P(e_J)
\end{aligned} \tag{11}
$$

Next, we derive the statistical model from the system's CPDs. This derivation is possible because a fully specified SCM encodes both functional and probabilistic information, which we can replace with suitable functions and distributional assumptions based on the available data (Pearl et al., 2016). Figure 9 illustrates the progression from the SCM to the probabilistic model and, ultimately, to the statistical model. The model starts by assuming that O_R follows a Bernoulli distribution[5], reflecting the binary nature of CJ outcomes. Furthermore, following the conventions of Generalized Linear Models (GLMs) (McCullagh and Nelder, 1983; Lee and Nelder, 1996; Agresti, 2015), the distribution links O_R to the latent discriminal difference vector D_R using an inverse-logit function: $\text{inv\_logit}(x) = 1/(1 + \exp(-x))$.

Next, D_R is defined as the difference between the discriminal processes $T_{IA}[i, a]$ and $T_{IA}[h, b]$, representing the underlying written-quality trait of the compared texts, plus the corresponding judgment-level bias $B_{JK}[j, k]$. Note that if it is assumed that $B_{JK}[j, k]$ reflects the difference in stimulus-specific biases, i.e., $B_{JK}[j, k] = B_{JK}[i, a, j, k] - B_{JK}[h, b, j, k]$, the discriminal difference can be re-written as:

$$
\begin{aligned}
D_R &= (T_{IA}[i, a] - T_{IA}[h, b]) + B_{JK}[j, k] \\
&= (T_{IA}[i, a] - T_{IA}[h, b]) + (B_{JK}[i, a, j, k] - B_{JK}[h, b, j, k]) \\
&= (T_{IA}[i, a] + B_{JK}[i, a, j, k]) - (T_{IA}[h, b] + B_{JK}[h, b, j, k]) \\
&= T^*_{IA}[i, a] - T^*_{IA}[h, b]
\end{aligned} \tag{12}
$$

This formulation reveals that the discriminal difference captures a *pure interaction effect*, in which neither the texts' discriminal processes nor the judges' biases alone determine the outcome, but

[5] The binomial distribution–including its special case, the Bernoulli distribution–represent a maximum entropy distribution for binary events (McElreath, 2020, p. 34). This means that the Bernoulli distribution is the most consistent alternative when only two un-ordered outcomes are possible and their expected frequencies are assumed to be constant (McElreath, 2020, p. 310). For a detailed discussion of the binomial as a maximum entropy distribution, see McElreath (2020, sec. 10.1.2).

$$
\begin{array}{lll}
O_R := f_O(D_R, S, C) & P(O_R \mid D_R) & O_R \overset{iid}{\sim} \text{Bernoulli}\left[\text{inv\_logit}(D_R)\right] \\
D_R := f_D(T_{IA}, B_{JK}) & P(D_R \mid T_{IA}, B_{JK}) & D_R = (T_{IA}[i,a] - T_{IA}[h,b]) + B_{JK}[j,k] \\
T_{IA} := f_T(T_I, X_{IA}, e_{IA}) & P(T_{IA} \mid T_I, X_{IA}, e_{IA}) & T_{IA} = T_I + \beta_{XA} X_{IA} + e_{IA} \\
T_I := f_T(X_I, e_I) & P(T_I \mid X_I, e_I) & T_I = \beta_{XI} X_I + e_I \\
B_{JK} := f_B(B_J, Z_{JK}, e_{JK}) & P(B_{JK} \mid B_J, Z_{JK}, e_{JK}) & B_{JK} = B_J + \beta_{ZK} Z_{JK} + e_{JK} \\
B_J := f_B(Z_J, e_J) & P(B_J \mid Z_J, e_J) & B_J = \beta_{ZJ} Z_J + e_J \\
 & & \\
e_I \perp \{e_J, e_{IA}, e_{JK}\} & P(e_I)P(e_{IA})P(e_J)P(e_{JK}) & e \sim \text{Multi-Normal}(\mu, \Sigma) \\
e_J \perp \{e_{IA}, e_{JK}\} & & \Sigma = VQV \\
e_{IA} \perp e_{JK} & &
\end{array}
$$

(a) SCM (b) Probabilistic model (c) Statistical model

Figure 9: Comparative judgment model. SCM, probabilistic and statistical model assuming different discriminal dispersions for the student's traits

their interaction does (Attia et al., 2022). Put simply, this mathematical description captures the idea that the stimuli' discriminal processes become an observable outcome only through the lens of judges' perceptions (i.e., their biases). For clarity, the square brackets in D_R indicate the relevant indices for each trait vector.

Now the functional forms for T_{IA}, T_I, B_{JK}, and B_J are specified. T_{IA} is modeled as a linear combination of the students' underlying writing-quality traits T_I, the effects of relevant text-related variables on quality assessment $\beta_{XA} X_{IA}$ (such as the influence of text length), and the text-specific idiosyncratic errors e_{IA}. Similarly, T_I is expressed as a linear combination of relevant student-related variables affecting the quality assessment $\beta_{XI} X_I$, and student-specific idiosyncratic errors e_I. For the judge-specific terms, B_{JK} is modeled as a linear combination of the judge's individual bias B_J, the influence of relevant judgment-related variables on quality assessment $\beta_{ZK} Z_{JK}$ (e.g., how the number of judgments affect the evaluation), and judgment-specific idiosyncratic errors e_{JK}. Finally, B_J is defined as a linear combination of relevant judge-level variables influencing the quality assessment $\beta_{ZJ} Z_J$ (such as judgment expertise) and judge-specific idiosyncratic errors e_J.

Next, the probabilistic assumptions for the idiosyncratic errors e_I, e_{IA}, e_J, and e_{JK} are specified. Unlike other variables in the model, these error terms exhibit indeterminacies in their *location*, *orientation*, and *scale* due to the lack of an inherent scale in the associated latent variables T_I, T_{IA}, B_J, and B_{JK}. Thus, to identify the latent variable model these indeterminacies must be resolved (Depaoli, 2021; de Ayala, 2009). Drawing on principles from SEM (Hoyle, 2023), the vector of idiosyncratic errors $e = [e_I, e_{IA}, e_J, e_{JK}]^T$ are assumed to follow a Multivariate Normal distribution with mean vector μ and a covariance matrix $\Sigma = VQV$, with V denoting a diagonal matrix of standard deviations and Q a correlation matrix. To address the *location* indeterminacy, the errors' mean vector is set to zero:

$$\mu = [0, 0, 0, 0]^T \tag{13}$$

Following SCM 9a, the *orientation* indeterminacy is solved by assuming that the errors are uncorrelated. This assumption leads to the definition of the error's correlation matrix, Q, as the identity matrix:

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{14}$$

To resolve the *scale* indeterminacy, the diagonal matrix V is defined as follows:

$$V = \begin{bmatrix} s_{XI} & 0 & 0 & 0 \\ 0 & p_{IA} & 0 & 0 \\ 0 & 0 & s_{ZJ} & 0 \\ 0 & 0 & 0 & p_{JK} \end{bmatrix} \tag{15}$$

Here, s_{XI} represents the standard deviation for the individuals, p_{IA} for the stimuli, s_{ZJ} for the judges, and p_{JK} for the judgments. It is assumed s_{XI} varies depending on the teaching method group to which each student belongs. Using the example from Section 4, where the teaching method $X_I = \{1, 2\}$, the model sets the constraint according to Equation 16. This constraint anchors the scale of the individuals' latent trait while relaxing the assumption of equal dispersion for the stimuli, thereby addressing the concerns raised in Section 3.1.1.

$$\sum_{g=1}^{2} s_{XI}[g]/2 = 1 \tag{16}$$

Because the error vector e follows an uncorrelated Multivariate Normal distribution, the marginal distribution of e_{IA} is a univariate Normal distribution with mean zero and standard deviation p_{IA}. Thus, p_{IA} is set as a proportion of 1 to establish the scale of the stimuli' latent trait relative to the scale of the individuals' trait. Note that as a result, T_{IA} is also normally distributed. This configuration effectively reinstates Thurstone's original assumption of Normal discriminal processes for the stimuli (see Table 1).

Similarly, it is assumed that s_{ZJ} varies depending on the groups to which each judge belongs. For instance, if $Z_J = \{1, 2, 3\}$ represents three groups of judges with varying expertise, the model sets the constraint according to Equation 17. This constraint anchors the scale of the judges' latent trait and relaxes the assumption of equal dispersion for the judgments.

$$\sum_{g=1}^{3} s_{ZJ}[g]/3 = 1 \tag{17}$$

Conversely, p_{JK} is defined as a proportion of 1 to establish the scale of the judgments' latent trait relative to the scale of the judges' trait.

Finally, we use *Bayesian inference methods* to convert the statistical model 9c into a practical statistical tool for analyzing paired comparison data. Bayesian inference offers three main advantages in this context. First, it handles complex and overparameterized models, where the number of parameters exceeds the number of observations (Baker, 1998; Kim and Cohen, 1999). This feature is essential for our implementation, as the proposed model is indeed overparameterized. Second, it incorporates prior information to constrain parameter estimates within plausible bounds, thereby mitigating estimation issues like non-convergence or improper solutions that often affect frequentist methods (Martin and McDonald, 1975; Seaman III et al., 2011). Prior distributions are used to define the error distribution and set the scale of latent variables (Depaoli, 2014). Third, Bayesian inference supports robust inferences from small samples, where the asymptotic properties underlying frequentist methods are less reliable (Baldwin and Fellingham, 2013; Lambert et al., 2006; Depaoli, 2014). This feature is particularly relevant in CJ assessments, as researchers often collect large volumes of paired comparisons but work with relatively small samples of judges, stimuli, and individuals to test hypotheses.

The **Declarations** section of this document provides a link to the model code, along with an alternative specification that assumes equal discriminal dispersions. Both versions of the model have been tested with success using `Stan` (version 2.26.1, Stan Development Team., 2026,?).

6. Discussion

Thurstone introduced the Law of Comparative Judgment to measure psychological traits of stimuli through pairwise comparisons (Thurstone, 1927a,b). In its general form, the theory models single-judge comparisons across multiple, potentially correlated stimuli. Each comparison produces a dichotomous outcome indicating which stimulus the judge perceives as having a higher trait level. However, Thurstone identified one key challenge in this general formulation: the measurement model required estimating more "unknown" parameters than the number of available pairwise comparisons (Thurstone, 1927b). To address this issue and to facilitate the theory's practical applicability, he formulated five cases, each progressively incorporating several simplifying assumptions.

Among these, Case V remains the most widely used model in empirical CJ research, mainly due to the widespread adoption of the BTL model. The BTL model incorporates the core assumptions of Case V–namely, equal discriminal dispersions and zero correlation among stimuli' discriminal processes–but replaces the processes' normal distribution with the more mathematically tractable logistic distribution (Andrich, 1978; Bramley, 2008). Although this substitution has minimal impact on trait estimation or model interpretation (van der Linden, 2017a; McElreath, 2021), the simplifying assumptions of the BTL model–and by extension, of Case V–may fail to capture the complexity of some traits or account for heterogeneous stimuli (Thurstone, 1927a; Andrich, 1978; Bramley, 2008; Kelly et al., 2022), potentially leading to unreliable and inaccurate trait estimates (Ackerman, 1989; Zimmerman, 1994; McElreath, 2020; Hoyle, 2023).

Moreover, because Thurstone's original goal was to produce a "coarse scaling" of traits and allocate stimuli along this continuum (Thurstone, 1927a, p. 269), his theory offered no guidance on how to use trait estimates for statistical inference. The CJ tradition has attempted to address this gap by separating trait estimation from hypothesis testing, relying on point estimates, such as BTL scores or their transformations, for inference. While this approach simplifies analysis, it can also introduce bias and compromise the reliability of the resulting inferences (McElreath, 2020; Kline, 2023; Hoyle, 2023).

To address the limitations of Thurstone's Case V and the BTL model, this study extended Thurstone's general form using a systematic, integrated approach that combined Causal and Bayesian inference methods. The approach began with the development of a conceptual model, formalized as a Structural Causal Model (SCM) and represented graphically by a Directed Acyclic Graph (DAG) (Pearl, 2009; Pearl et al., 2016; Gross et al., 2018; Neal, 2020). This model integrated Thurstone's core theoretical principles, such as the discriminal processes of stimuli, alongside key CJ assessment design features, including judges' bias, sampling procedures, and comparison mechanisms. Together, these components allowed the causal processes underlying the CJ system to be disentangled.

The approach then translated the SCM into a bespoke statistical model that enabled the analysis of CJ data in cases where the assumptions of equal dispersion and zero correlation were violated, and where statistical inference was required. In particular, the model accounted for judge biases, captured the hierarchical structure of stimuli, incorporated measurement error into the hypothesis testing process, and accommodated heterogeneity in discriminal dispersions. By addressing these issues, the methodological innovations aimed to enhance the reliability and validity of trait measurement in CJ, while also improving the accuracy of statistical inferences.

Beyond these potential benefits, the approach offered two additional advantages. First, it clarified the roles and interactions of all actors and processes involved in CJ assessments. Second, it shifted the analytic paradigm from passively accepting the assumptions of Case V and the BTL model to actively testing their fit with observed data. Together, these advantages established a principled framework for evaluating best practices in designing CJ assessments–one that better aligns with the demands of contemporary CJ contexts (Kelly et al., 2022)–providing new insights into existing research and opening promising avenues for future inquiry.

6.1. *Future research directions using our approach*

Among the many potential directions for future research, three avenues deserve particular attention due to their direct impact on the reliability and validity of CJ trait estimates, as well as on the accuracy of statistical inferences. The following sections outline these avenues and explain how our approach facilitates their investigation.

6.1.1. *The impact of sampling and comparison mechanisms*

Although sampling and comparison mechanisms are central to contemporary CJ assessments, it is striking that most CJ literature has examined them within a limited scope. Researchers have primarily investigated the effects of adaptive comparative judgment (ACJ) designs on trait reliability (Pollitt, 2012a,b; Bramley, 2015; Verhavert et al., 2022; Mikhailiuk et al., 2021) or proposed practical guidelines for the number of comparisons judges should make (Verhavert et al., 2019; Crompvoets et al., 2022). While these studies offer valuable insights, they also overlook the broader role that these mechanisms play within the CJ system. As this oversight likely stems from a more fundamental lack of conceptual clarity about how these mechanisms function within the system, the present study integrated these mechanisms into the conceptual model of CJ.

The explicit integration of the sampling and comparison mechanisms offers a new perspective on how these mechanisms shape the CJ process. Specifically, it clarified their role as sources of missing data in CJ's data-generating process–that is, as mechanisms that determine which observations are missing from the final data sample (Kohler et al., 2019; McElreath, 2020, chap. 15; Schuessler and Selb, 2023). This new perspective invites the application of Little and Rubin's principled missing data framework (Little and Rubin, 2020), allowing a more rigorous evaluation of existing claims about these missing data mechanisms, their influence on CJ outcomes, and their implications for designing and evaluating more complex assessments setups.

This study circumvented the need to apply the missing data framework by deliberately structuring the sampling and comparison mechanisms to be independent of any observed or unobserved variables, including the outcome. In other words, these mechanisms were intentionally designed to yield data that are *missing completely at random* (MCAR) (Little and Rubin, 2020). This design choice offered one key advantage: it generated simple random samples that satisfied the condition of *ignorability*, thereby allowing researchers to legitimately *ignore* missing data during analysis without introducing bias (Everitt and Skrondal, 2010; Kohler et al., 2019; Neal, 2020).

However, many modern CJ applications rely on more complex assessment designs, in which the sampling and comparison mechanisms introduce more intricate forms of missingness such as *missing* *at random* (MAR) or *missing not at random* (MNAR) (Little and Rubin, 2020). One prominent example is ACJ designs, where prior judgment outcomes inform the selection of stimulus pairs for subsequent comparisons (Pollitt, 2012a,b; Bramley, 2015). This pair selection procedure suggests that ACJ's comparison mechanism is outcome-dependent, potentially classifying the method as a generator of MNAR data. If this classification holds, ACJ might violate the condition of ignorability, making it an unsuitable pair selection procedure for reliable and valid trait estimation and inference. Moreover, under this interpretation, the mixed findings on ACJ's effectiveness become more comprehensible: while some studies find that the method improves trait reliability (Pollitt and Elliott, 2003; Pollitt, 2012a,b), others argue that that these gains may be artificially inflated (Bramley, 2015; Bramley and Vitello, 2019; Crompvoets et al., 2020, 2022).

Regardless of the underlying missingness mechanisms, any CJ assessment design would benefit from explicitly defining its assumptions–a practice supported by this study's approach. This clarity enables a thorough evaluation of how the sampling and comparison mechanisms affect trait estimation and statistical inference in each design. Such assessments are particularly relevant given the common misconception in the CJ literature that Thurstone's Case V and the BTL models can naturally handle even non-random missing data without compromising the reliability or validity of trait estimates (Bramley, 2008).

6.1.2. The effects of judges' bias on the reliability of trait estimates

Despite the growing notion that various stimulus-related factors influence judges' perceptions (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022) and that these influences may not always cancel out, empirical evidence of judges' biases remains scarce in the CJ literature (Pollitt and Elliott, 2003; van Daal et al., 2016; Bartholomew et al., 2020). This gap likely persists not from a lack of interest or research but because the identification of such biases often depends on ad-hoc detection methods, like 'misfit' statistics, that may not be well-suited for the task see, 3.2.

To overcome this limitation, the present study treated judges' biases as an integral component of the CJ system from the outset. This approach offers one key advantage: it provides a more accurate representation of the data-generating process behind pairwise comparisons, one that acknowledges that the discriminal processes of stimuli becomes an observable outcome only through judges' perceptions, which may exhibit bias.

The explicit integration of judges' bias into CJ's conceptual model then paves the way for investigating several relevant research questions. One key question is whether CJ data can be validly analyzed under the assumption of "sample-free" trait calibration, specifically under the hypothesis that judges exhibit no systematic bias . This question is particularly relevant because "sample-freeness" is still regarded as an inherent property of the BTL model (Bramley, 2008; Andrich, 1978) despite growing evidence of persistent biases. Another critical question is whether training or expertise can help judges avoid focusing on irrelevant stimulus features, such as handwriting, over more central criteria like argumentative quality in writing assessments (Kelly et al., 2022). Exploring these questions may also provide insights into what it truly means to be an "expert" within the CJ context (Kelly et al., 2022).

Moreover, since judges rely on these stimulus-related factors when evaluating complex, multidimensional traits (e.g., structure, style) (van Daal et al., 2016; Lesterhuis et al., 2018; Chambers and Cunningham, 2022), and these factors account for variation in judgment accuracy (Gill and Bramley, 2013; van Daal et al., 2017; van Daal, 2020; Gijsen et al., 2021), it is reasonable to expect that assessments also vary according to judge-specific attributes such as gender, age, culture, income, education, training, or expertise (Kelly et al., 2022). Prior studies support this view (e.g., Bartholomew et al., 2020; McMahon and Jones, 2015). Thus, building on the discussion in Section 6.1.1, further exploration is warranted into how judge selection influences the formation of a "shared consensus" and whether certain judge attributes introduce systematic biases or distortions in the stimuli trait distribution (Deffner et al., 2022). Furthermore, if such attributes do in fact compromise the assumption of "sample-freeness," it becomes essential to consider strategies for mitigating their effects and to determine how many judges (and how many judgments per judge) are required to produce reliable trait estimates under these conditions. Additionally, it is worth examining whether *repeated measures designs*, in which judges evaluate the same stimulus pairs multiple times (Lawson, 2015), can improve judgment consistency and accuracy. As anticipated, the approach presented in this study provides a structured foundation to rigorously investigate these questions.

6.1.3. The identification of 'misfitting' judges and stimuli

Although the CJ literature clearly defines *misfit* judges and stimuli, CJ researchers have rarely examined how these observations relate to Thurstonian theory. In particular, they have not identified which elements of Thurstone's theory account for the occurrence of misfits. This disconnect likely stems from the fact that misfit statistics are derived from residual analysis and outlier detection methods rather than from Thurstonian principles. Specifically, *misfit judges* are typically defined as those whose assessments diverge significantly from the "shared consensus" (Pollitt, 2012a,b; van Daal et al., 2016; Goossens and De Maeyer, 2018; Wu et al., 2022), while *misfit stimuli* are those that elicit more judgment discrepancies than others (Pollitt, 2004, 2012a,b; Goossens and De Maeyer, 2018). Both definitions closely mirror the statistical concept of outliers, that is, observations that deviate markedly from the rest of the sample in which they occur (Grubbs, 1969). But this resemblance extends beyond the definitions themselves, as misfits are often identified using conventional outlier detection procedures, such as transforming BTL model residuals into diagnostic statistics and comparing them against predefined thresholds (Pollitt, 2012a,b; Wu et al., 2022).

Nevertheless, is not the classification of misfits as outliers that raises concerns for trait measurement and inference. Rather, the concern lies in the prevailing CJ practice of identifying these observations through ad-hoc procedures and excluding them from analysis (Pollitt, 2012a,b), often without empirical support for the various hypotheses suggested to explain their occurrence. It is essential to recognize that outliers are defined relative to a specific model (McElreath, 2020). Thus, detection procedures based on models like the BTL–which rests on strong assumptions–should be approached with caution, as they may not be appropriate for this purpose (Kelly et al., 2022). If the BTL model does not accurately represent the actual data-generating process of the CJ system, the method may miss classify valid observations as misfits. On top this, excluding these observations carries additional risks. The statistical literature cautions that removing outliers can discard valuable information (Miller, 2023) and introduce bias into trait estimates. The direction and magnitude of this bias are often unpredictable, as they depend on which observations are excluded from the analysis (Zimmerman, 1994; O'Hagan, 2018; McElreath, 2020). Finally, even when the model aligns well with the data, its rigid assumptions limit the model's ability to adequately test many of the hypotheses proposed to explain misfit behavior.

In contrast, the approach presented in this study provides a rigorous framework for testing several relevant hypotheses. For instance, it allows to investigate whether misfit judges are those who exhibit (an outlying degree of) systematic bias or greater variability in their judgments compared to their peers. Similarly, the approach makes possible to explore whether misfit stimuli exhibit more variable discriminal processes relative to other stimuli or if they are genuinely outlying cases. Moreover, since outliers are not inherently "bad data" (McElreath, 2020), the present approach offers a principled alternative to exclusion, one that retains misfits in the analysis without compromising trait estimation or inference. This is achieved by adapting the proposed model into robust measurement models (McElreath, 2020), a broad class of procedures designed to reduce the sensitivity of parameter estimates to mild or moderate departures from model assumptions (Everitt and Skrondal, 2010). This strategy also speaks to a broader concern in the social sciences: when predictive power is low, the solution may not lie solely in seeking new variables or procedures, but also in adopting more sophisticated measurement models (Wainer et al., 1978).

6.1.4. The outcome of analysis

6.2. Study limitations and practical challenges for applied CJ researchers

Drawing conclusions from observed data always requires assumptions, whether the data are observational or experimental (Kohler et al., 2019; Deffner et al., 2022). The proposed approach is not an exception to this fundamental principle. Like all approaches grounded on Causal inference, it relies on expert knowledge and assumptions about the variables' causal structure that are often untestable at the outset (Hernán and Robins, 2025). However, its purpose is not to deliver automatic answers when applied to a given CJ dataset. Instead, it encourages the formulation of precise questions and the explicit articulation of assumptions, fostering a generalizable understanding of the CJ system under study (Rohrer et al., 2022; Deffner et al., 2022; Sterner et al., 2024). This clarity is crucial because the accuracy of estimates and validity of inferences depend on how well the data and inferential goals align with a model's assumptions (Kohler et al., 2019). Although this alignment remains to be empirically tested for the proposed statistical model, the theory-driven nature of this approach provides a solid foundation for future empirical evaluation of its causal assumptions (Deffner et al., 2022), grounded in both established theory and existing evidence.

The theoretical commitment to causal inference also introduces several practical challenges that applied CJ researchers must navigate. These fall into two main categories: first, acquiring the foundational knowledge necessary to apply the approach effectively; and second, dedicating greater attention to both conceptual and statistical modeling.

6.2.1. Required foundational knowledge

Applying the approach effectively requires foundational knowledge in two areas. First, a solid understanding of Causal inference principles. Second, the ability to translate the functional and probabilistic aspects of conceptual models into bespoke statistical models. A clear example illustrating the importance of Causal inference is the recurrent assumption that predictor variables are "relevant" to the research context–interpreting this relevance as their inclusion in a *sufficient adjustment set* (Pearl, 2009; Pearl et al., 2016; Morgan and Winship, 2014). However, this study does not explore the full implications of that assumption for model specification, estimation, and inference. Nevertheless, to assist with the effort of engaging with this and other complex Causal inference concepts, key references are provided throughout the study to guide applied CJ researchers toward a deeper understanding of these ideas [6].

Developing the skills to translate conceptual models into bespoke statistical models also presents challenges. Although Bayesian inference methods offer a more accessible path for many applied CJ researchers to develop these skills–by reducing the need for specialized knowledge in areas like optimization theory, which frequentist methods often require–they still demand familiarity with technical concepts such as probabilistic programming languages (PPLs), probability distributions, and convergence diagnostics. Thus, to support this skill development, this study provides a link to the statistical model code and alternative model specifications in the **Declarations** section of this document [7].

6.2.2. Attention to conceptual and statistical modeling

Even after acquiring the necessary foundational knowledge, applying this approach to a specific CJ context involves two additional challenges. First, it is essential to verify whether the conceptual

[6]Also refer to the detailed online document referenced in the **Declarations** section

[7]Seminal texts on Bayesian inference methods, such as Gelman et al. (2014) and McElreath (2020), offer valuable support for developing a deeper understanding of these models.

model achieves sufficient *fidelity* in representing the CJ system under study. Second, the statistical translation of the model must be assessed to determine if it can accurately estimate the intended estimands (i.e., parameters) from empirical data. To ensure fidelity, the conceptual and statistical models presented here should be treated as starting points, not universal solutions for all CJ designs or datasets. While these models may be adequate in some contexts, assuming so without evaluation is unwise. Instead, adaptation to the specific context is necessary, with careful attention to assessment design features and their implied causal assumptions. As discussed in Section 6.2, this approach supports such adaptation by providing a transparent framework for articulating new assumptions and guiding CJ assessment design.

Conversely, evaluating the estimation capabilities of the statistical model requires addressing the challenge of identification analysis. As outlined in Section 4, *identification analysis* determines whether a statistical model can accurately compute a given estimand (e.g., a parameter) based solely on its (causal) assumptions, independent of random variability (Schuessler and Selb, 2023). Identification is crucial because it is a necessary condition for consistency. *Consistency* is the property of an estimator (e.g., a statistical model) whose estimates converge to the "true" value of an estimand as the data size approaches infinity (Everitt and Skrondal, 2010). Without identification, consistency is impossible–even with infinite, error-free data–and meaningful inference from finite samples cannot be achieved (Schuessler and Selb, 2023). While formal derivations of identification may seem a natural next step, the complexity of the CJ system makes this approach impractical at the outset. Instead, simulation-based methods like power analysis offer a more practical and flexible alternative, enabling examination of estimate consistency without relying on complex mathematical proofs. Notably, the approach presented here supports both strategies by providing the probabilistic foundation for formal derivations and the statistical structure needed for simulation-based methods.

7. Conclusion

The present study highlights the need to extend Thurstone's theory to address the demands of contemporary empirical CJ research. It advocates for developing bespoke CJ models tailored to the specific data-generating processes of different CJ assessment designs. These models aim to enhance the robustness of trait estimates and clarity of inferences. Moreover, this work outlines a clear path for advancing both theoretical and applied CJ research and lays the foundation for broader adoption of more robust and interpretable methodologies across the social sciences.

Declarations

Funding: The Research Fund (BOF) of the University of Antwerp funded this project.

Financial interests: The authors declare no relevant financial interests.

Non-financial interests: The authors declare no relevant non-financial interests.

Ethics approval: The University of Antwerp Research Ethics Committee confirmed that this study does not require ethical approval.

Consent to participate: Not applicable

Consent for publication: All authors have read and approved the final version of the manuscript for publication.

Data availability: This study did not use any data.

Materials and code availability: A previous version of this manuscript, along with the associated materials and code (see the section titled `CODE LINK`), has been made publicly available at: https://jriveraespejo.github.io/paper2_manuscript/.

AI-assisted technologies in the writing process: The authors used various AI-based language tools to refine phrasing, optimize wording, and enhance clarity and coherence throughout the manuscript. They take full responsibility for the final content of the publication.

CRediT authorship contribution statement: *Conceptualization:* J.M.R.E, T.vD., S.DM., and S.G.; *Methodology:* J.M.R.E, T.vD., and S.DM.; *Software:* J.M.R.E.; *Validation:* J.M.R.E.; *Formal Analysis:* J.M.R.E.; *Investigation:* J.M.R.E; *Resources:* T.vD. and S.DM.; *Data curation:* J.M.R.E.; *Writing - original draft:* J.M.R.E.; *Writing - review and editing:* J.M.R.E., T.vD., S.DM., and S.G.; *Visualization:* J.M.R.E.; *Supervision:* S.G. and S.DM.; *Project administration:* S.G. and S.DM.; *Funding acquisition:* S.G. and S.DM.

8. Appendix

8.1. Statistical and Causal inference

This section introduces fundamental statistical and Causal inference concepts necessary for understanding the core theoretical principles described in this document. It does not, however, offer a comprehensive overview of Causal inference methods. Readers seeking more in-depth understanding may wish to explore introductory papers such as Pearl (2010), Rohrer (2018), Pearl (2019), and Cinelli et al. (2020). They may also find it helpful to consult introductory books like Pearl and Mackenzie (2018), Neal (2020), and McElreath (2020). For more advanced study, readers may refer to seminal intermediate papers such as Neyman (1923), Rubin (1974), Spirtes et al. (1991), and Sekhon (2009), as well as books such as Pearl (2009), Morgan and Winship (2014), and Hernán and Robins (2025).

8.1.1. Empirical research and randomized experiments

Empirical research uses evidence from observation and experimentation to address real-world challenges. In this context, researchers typically formulate their research questions as *estimands* or *targets of inference*, i.e., the specific quantities they seek to determine (Everitt and Skrondal, 2010). For instance, researchers might be interested in answering the following question: "To what extent do different teaching methods (T) influence students' ability to produce high-quality written texts (Y)?" To investigate this, researchers could randomly assign students to two groups, each exposed to a different teaching method ($T_i = \{1, 2\}$). Then, they would perform pairwise comparisons, generating a dichotomous outcome ($Y_i = \{0, 1\}$) showing which student exhibits more of the ability. In this scenario, the research question can be rephrased as the estimand, "*On average*, is there a difference in the ability to produce high-quality written texts between the two groups of students?" and this estimand can be mathematically represented by the random associational quantity in Equation 18, where $E[\cdot]$ denotes the expected value.

$$E[Y_i \mid T_i = 1] - E[Y_i \mid T_i = 2] \tag{18}$$

Researchers then proceed to identify the estimands. *Identification* determines whether an estimator can accurately compute the estimand based solely on its assumptions, regardless of random variability (Schuessler and Selb, 2023, p. 4). An *estimator* refers to a method or function that transforms data into an estimate (Neal, 2020). *Estimates* are numerical values that approximate the estimand derived through the process of *estimation*, which integrates data with an estimator (Everitt and Skrondal, 2010). The Identification-Estimation flowchart (McElreath, 2020; Neal, 2020), shown in Figure 10, visually represents the transition from estimands to estimates.

Identification is a necessary condition to ensure *consistent* estimators. An estimator achieves *consistency* when it converges to the "true" value of an estimand as the data size approaches infinity

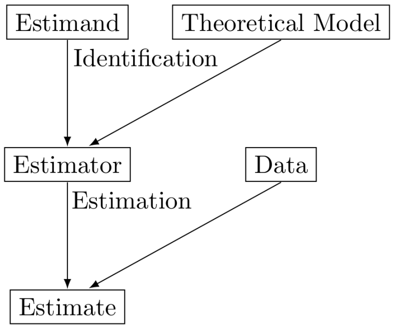


Figure 10: Identification-Estimation flowchart. Extracted and slightly modified from Neal (2020, p. 32)

(Everitt and Skrondal, 2010). Without identification, researchers cannot achieve consistency, even with "infinite" and error-free data. As a result, deriving meaningful insights about an estimand from finite data becomes impossible (Schuessler and Selb, 2023, p. 5). Therefore, to ensure accurate and reliable estimates, researchers prioritize estimators with desirable identification properties. For instance, the Z-test is a widely used estimator for comparing group proportions, yielding accurate estimates when its underlying assumptions are satisfied (Kanji, 2006). Furthermore, researchers can interpret estimates from the Z-test as causal, provided the data is collected through a randomized experiment.

Randomized experiments are widely recognized as the gold standard in evidence-based science (Hariton and Locascio, 2018; Hansson, 2014). This recognition stems from their ability to enable researchers interpret associational estimates as causal. They achieve this by ensuring data, and by extension an estimator, satisfies several key identification properties, such as common support, no interference, and consistency (Morgan and Winship, 2014; Neal, 2020). The most critical property, however, is the elimination of confounding. *Confounding* occurs when an external variable X simultaneously influences the outcome Y and the variable of interest T, resulting in spurious associations (Everitt and Skrondal, 2010). Randomization addresses this issue by decoupling the association between the intervention allocation T from any other variable X (Morgan and Winship, 2014; Neal, 2020).

Nevertheless, researchers often face constraints that limit their ability to conduct randomized experiments. These constraints include ethical concerns, such as the assignment of individuals to potentially harmful interventions, and practical limitations, such as the infeasibility of, for example, assigning individuals to genetic modifications or physical impairments (Neal, 2020). In these cases, Causal inference offers a valuable alternative for generating causal estimates and understanding the mechanisms underlying specific data. In addition, the framework can provide significant theoretical insights that can enhance the design of experimental and observational studies (McElreath, 2020).

8.1.2. Identification under Causal inference

Unlike classical statistical modeling, which focuses primarily on summarizing data and inferring associations, the *Causal inference* framework is designed to identify causes and estimate their effects using data (Shaughnessy et al., 2010; Neal, 2020). The framework uses rigorous mathematical techniques to address the *fundamental problem of causality* (Pearl, 2009; Pearl et al., 2016; Morgan and Winship, 2014). This problem revolves around the question, "What would have happened 'in the world' under different circumstances?" This question introduces the concept of counterfactuals, which are instrumental in defining and identifying causal effects.

Counterfactuals are hypothetical scenarios that are *contrary to fact*, where alternative outcomes resulting from a given cause are neither observed nor observable (Neal, 2020; Counterfactual, 2024). The structural approach to Causal inference (Pearl, 2009; Pearl et al., 2016) provides a formal framework for defining counterfactuals. For instance, in the scenario described in Section 8.1.1, the approach begins by defining the *individual causal effect* (ICE) as the difference between each student's potential outcomes, as in Equation 19.

$$\tau_i = Y_i \mid do(T_i = 1) - Y_i \mid do(T_i = 2) \tag{19}$$

where $do(T_i = t)$ represents the intervention operator, $Y_i \mid do(T_i = 1)$ represents the potential outcome under intervention $T_i = 1$, and $Y_i \mid do(T_i = 1)$ represents the potential outcome under intervention $T_i = 2$. Here, an *intervention* involves assigning a constant value to the treatment variable for each student's potential outcomes. Note that if a student is assigned to intervention $T_i = 1$, the potential outcome under $T_i = 2$ becomes a counterfactual, as it is no longer observed nor observable. To address this challenge, the structural approach extends the ICE to the *average causal effect* (ACE, Equation 20), representing the average difference between the students' observed potential outcomes and their counterfactual counterparts.

$$\begin{aligned} \tau &= E[\tau_i] \\ &= E[Y_i \mid do(T_i = 1)] - E[Y_i \mid do(T_i = 2)] \end{aligned} \tag{20}$$

Even though counterfactuals are unobservable, researchers can still identify the ACE from associational estimates by leveraging the structural approach. The approach identifies the ACE by statistically conditioning data on a *sufficient adjustment set* of variables X (Pearl, 2009; Pearl et al., 2016; Morgan and Winship, 2014). This *sufficient* set (potentially empty) must block all non-causal paths between T to Y without opening new ones. When such a set exists, then T and Y are *d-separated* by X ($T \perp Y \mid X$) (Pearl, 2009), and X satisfies the *backdoor criterion* (Neal, 2020, pp 37). Here, *conditioning* describes the process of restricting the focus to the subset of the population defined by the conditioning variable (Neal, 2020, p. 32) (see Equation 21).

Conditioning on a sufficient adjustment set enables researchers to estimate the ACE, even when the data comes from an observational study. This process is feasible because such conditioning ensures that the ACE estimator satisfies several critical properties, including confounding elimination (Morgan and Winship, 2014). Naturally, the validity of claims about the causal effects of T on Y now hinges on the assumption that X serves as a sufficient adjustment set. However, as Kohler et al. (2019, p. 150) noted, drawing conclusions about the real world from observed data inevitably requires assumptions. This requirement holds true for both observational and experimental data.

For instance, if researchers cannot conduct the randomized experiments described in Section 8.1.1 and must instead rely on observational data, they can still identify the ACE as long as an observed variable X, such as the socio-economic status of the school, satisfies the backdoor criterion. Under these circumstances, researchers first identify the *conditional average causal effect* (CACE, Equation 21)

$$CACE_t = E[Y_i \mid T_i = t, X] \tag{21}$$

From the CACE, researchers can identify the ACE from associational quantities as in Equation 22. This identification process is commonly known as the *backdoor adjustment*. Here, $E_X[\cdot]$ represents the marginal expected value over X (Morgan and Winship, 2014).

$$\begin{aligned} \tau &= E[Y_i \mid do(T_i = 1)] - E[Y_i \mid do(T_i = 2)] \\ &= E_X[CACE_1 - CACE_2] \\ &= E_X\left[E[Y_i \mid T_i = 1, X] - E[Y_i \mid T_i = 2, X]\right] \end{aligned} \tag{22}$$

Notably, the approach extends the ACE identification for a continuous variable T as in Equation 23, ensuring broad applicability across different causal scenarios (Neal, 2020, p. 45)

$$\begin{aligned} \tau &= E[Y_i \mid do(T_i = t)] \\ &= dE_X\left[E[Y_i \mid T_i = t, X]\right]/dt \end{aligned} \tag{23}$$

8.1.3. Diving into the specifics

The structural approach to Causal inference uses SCMs and DAGs to formally and graphically represent the presumed causal structure underlying the ACE (Pearl, 2009; Pearl et al., 2016; Gross et al., 2018; Neal, 2020). Essentially, these tools serve as *conceptual (theoretical) models* on which identification analysis rests (Schuessler and Selb, 2023, p. 4). Thus, using these tools, researchers can determine which statistical models can identify (ACE, CACE, or other), assuming the depicted causal structure is correct (McElreath, 2020), thus enabling valid Causal inference. Figure 10 shows the role of theoretical models in the inference process.

SCMs and DAGs support identification analysis through two key advantages. First, regardless of complexity, they can represent various causal structures using only five fundamental building blocks (Neal, 2020; McElreath, 2020). This feature allows researchers to decompose complex structures into manageable components, facilitating their analysis (McElreath, 2020). Second, they depict causal relationships in a non-parametric and fully interactive way. This flexibility enables feasible ACE identification strategies without defining the variables' data types, the functional form between them, or their parameters (Pearl et al., 2016, p. 35).

Thus, Section 8.1.3.1 and Section 8.1.3.2 elaborate on the first advantage, while Section 8.1.3.2 and Section 8.1.3.3 do so for the second. Finally, Section 8.1.3.4 explains how researchers use SCMs and DAGs alongside Bayesian inference methods in the estimation process.

8.1.3.1. The five fundamental block for SCMs and DAGs.

Figures 11, 12, 13, 14, and 15 display the five fundamental building blocks for SCMs and DAGs. The left panels of the figures show the formal mathematical models, represented by the SCMs, defined in terms of a set of *endogenous* variables $V = \{X_1, X_2, X_3\}$, a set of *exogenous* variables $E = \{e_{X1}, e_{X2}, e_{X3}\}$, and a set of functions $F = \{f_{X1}, f_{X2}, f_{X3}\}$ (Pearl, 2009; Cinelli et al., 2020). Endogenous variables are those whose causal mechanisms a researcher chooses to model (Neal, 2020). In contrast, exogenous variables represent *errors* or *disturbances* arising from omitted factors that the investigator chooses not to model explicitly (Pearl, 2009, p. 27,68). Lastly, the functions, referred to as *structural equations*, express the endogenous variables as non-parametric functions of other variables. These functions use the symbol '$:=$' to denote the asymmetrical causal dependence of the variables and the symbol '$\perp$' to represent *d-separation*, a concept akin to (conditional) independence.

Notably, every SCM has an associated DAG (Pearl et al., 2016; Cinelli et al., 2020). The right panels of the figures display these DAGs. A DAG is a graph consisting of nodes connected by edges, where the nodes represent random variables. The term *directed* means that the edges extend from one node to another, with arrows indicating the direction of causal influence. The term *acyclic* implies that the causal influences do not form loops, ensuring the influences do not cycle back on themselves (McElreath, 2020). DAGs represent observed variables as solid black circles, while they use open circles for unobserved (latent) variables (Morgan and Winship, 2014). Although the *standard representation* of DAGs typically omits exogenous variables for simplicity, the *magnified representation* depicted in the figures offers one key advantage: including exogenous variables can help researchers highlight potential issues related to conditioning and confounding (Cinelli et al., 2020).

(b) DAG

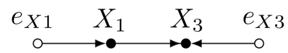
$$X_1 := f_{X1}(e_{X1})$$

$$X_3 := f_{X3}(e_{X3})$$

$$e_{X1} \perp e_{X3}$$

(a) SCM

Figure 11: Two unconnected nodes



(b) DAG

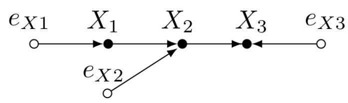
$$X_1 := f_{X1}(e_{X1})$$

$$X_3 := f_{X3}(X_1, e_{X3})$$

$$e_{X1} \perp e_{X3}$$

(a) SCM

Figure 12: Two connected nodes or descendant



(b) DAG

$$X_1 := f_{X1}(e_{X1})$$

$$X_2 := f_{X2}(X_1, e_{X2})$$

$$X_3 := f_{X3}(X_2, e_{X3})$$

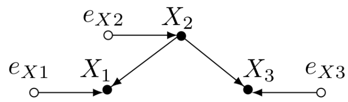
$$e_{X1} \perp e_{X2}$$

$$e_{X1} \perp e_{X3}$$

$$e_{X2} \perp e_{X3}$$

(a) SCM

Figure 13: Chain or mediator

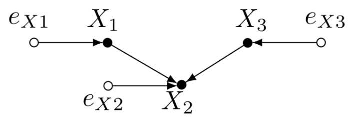


(b) DAG

$$
\begin{aligned}
X_1 &:= f_{X1}(X_2, e_{X1}) \\
X_2 &:= f_{X2}(e_{X2}) \\
X_3 &:= f_{X3}(X_2, e_{X3}) \\
e_{X1} &\perp e_{X2} \\
e_{X1} &\perp e_{X3} \\
e_{X2} &\perp e_{X3}
\end{aligned}
$$

(a) SCM

Figure 14: Fork or confounder



(b) DAG

$$
\begin{aligned}
X_1 &:= f_{X1}(e_{X1}) \\
X_2 &:= f_{X2}(X_1, X_3, e_{X2}) \\
X_3 &:= f_{X3}(e_{X3}) \\
e_{X1} &\perp e_{X2} \\
e_{X1} &\perp e_{X3} \\
e_{X2} &\perp e_{X3}
\end{aligned}
$$

(a) SCM

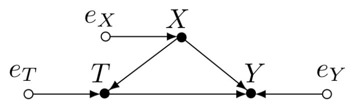
Figure 15: Collider or inmorality

A careful examination of these building blocks highlights the theoretical assumptions underlying their observed variables. SCM 11a and DAG 11b depict two unconnected nodes, representing a scenario where variables X_1 and X_3 are independent or not causally related. SCM 12a and DAG 12b illustrate two connected nodes, representing a scenario where a *parent* node X_1 exerts a causal influence on a *child* node X_3. In this setup, X_3 is considered a *descendant* of X_1. Additionally, X_1 and X_3 are described as *adjacent* because there is a *direct path* connecting them. SCM 13a and DAG 13b depict a *chain*, where X_1 influences X_2, and X_2 influences X_3. In this configuration, X_1 is a parent node of X_2, which is a parent node of X_3. This structure creates a *directed path* between X_1 and X_3. Consequently, X_1 is an *ancestor* of X_3, and X_2 fully *mediates* the relationship between the two. SCM 14a and DAG 14b illustrate a *fork*, where variables X_1 and X_3 are both influenced by X_2. Here, X_2 is a parent node that *confounds* the relationship between X_1 and X_3. Finally, SCM 15a and DAG 15b show a *collider*, where variables X_1 and X_3 are concurrent causes of X_2. In this configuration, X_1 and X_3 are not causally related to each other but both influence X_2 (an *inmorality*). Notably, all building blocks assume the errors are independent of each other and from all other variables in the graph, as evidenced by the pairwise relations $e_{X1} \perp e_{X2}$, $e_{X1} \perp e_{X3}$, and $e_{X2} \perp e_{X3}$.

Researchers can then use these building blocks to represent the scenario outlined in Section 8.1.2. SCM 16a and DAG 16b depict the plausible causal structure for this example. In this context, the variable X (socio-economic status of the school) is thought to be a confounder in the relationship between the teaching method T and the outcome Y. The figures display multiple descendant relationships such as $X \rightarrow T$, $X \rightarrow Y$, and $T \rightarrow Y$. They also highlight unconnected node pairs, evident from the relationships $e_T \perp e_X$, $e_T \perp e_Y$, and $e_X \perp e_Y$. Additional, the figures show one fork, $X \rightarrow \{T, Y\}$, and two colliders: $\{X, e_T\} \rightarrow T$ and $\{X, T, e_Y\} \rightarrow Y$.

$$
\begin{aligned}
X &:= f_X(e_X) \\
T &:= f_T(X, e_T) \\
Y &:= f_Y(T, X, e_Y) \\
e_T &\perp e_X \\
e_T &\perp e_Y \\
e_X &\perp e_Y
\end{aligned}
$$

(a) SCM



(b) DAG

Figure 16: Plausible causal structure the scenario outlined in Section 8.1.2.

8.1.3.2. The probabilistic implications of these blocks.

Beyond their graphical capabilities, researchers can translate SCMs and DAGs into statistical models because these conceptual models encode functional and probabilistic information, which they can then replace with suitable functions and probabilistic assumptions (Pearl et al., 2016). Notably, this encoding relies on two fundamental assumptions: the causal Markov and the faithfulness assumption. The *causal Markov assumption* links the graph structure to the joint probability distribution of variables by asserting that any variable in a graph is independent of all its non-descendants conditional on its parents (Neal, 2020, p. 20; Hernán and Robins, 2025, p. 80). The *faithfulness assumption*, in turn, states that the observed statistical (in)dependencies in the data reflect the underlying causal structure among variables–except in rare degenerate cases (Neal, 2020, p. 100; Hernán and Robins, 2025, p. 81).

A notable implication of the assumptions underlying the probabilistic encoding is that any conceptual model described by an SCM and DAG can represent the joint distribution of variables more efficiently (Pearl et al., 2016, p. 29). This expression takes the form of a product of conditional probability distributions (CPDs) of the type $P(child \mid parents)$. This property is formally known as the *Bayesian Network factorization* (BNF, Equation 24) (Pearl et al., 2016, p. 29; Neal, 2020, p. 21). In this expression, $pa(X_i)$ denotes the set of variables that are the parents of X_i.

$$\begin{aligned} P(X_1, X_2, \dots, X_P) &= X_1 \cdot \prod_{p=2}^{P} P(X_i \mid X_{i-1}, \dots, X_1) \quad &\text{(by chain rule)} \\ &= X_1 \cdot \prod_{p=2}^{P} P(X_i \mid pa(X_i)) \quad &\text{(by BNF)} \end{aligned} \tag{24}$$

This encoding enables researchers with conceptual (theoretical) knowledge in the form of an SCM and DAG to predict patterns of (in)dependencies in the data. As highlighted by Pearl et al. (2016, p. 35), these predictions depend solely on the structure of these conceptual models without requiring the quantitative details of the equations or the distributions of the errors. Moreover, once researchers observe empirical data, the patterns of (in)dependencies in the data can provide significant insights into the validity of the proposed conceptual model.

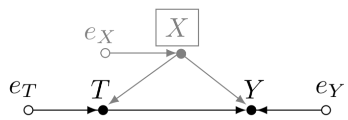
The five fundamental building blocks described in Section 8.1.3.1 clearly illustrate which (in)dependencies can SMCs and DAGs predict. For instance, applying the BNF to the causal structure shown in the SCM 11a and DAG 11b enables researchers to express the joint probability distribution of the observed variables as $P(X_1, X_3) = P(X_1)P(X_3)$, supporting the theoretical assumption that the observed variables X_1 and X_3 are unconditionally independent ($X_1 \perp X_3$) (Neal, 2020, p. 24). Conversely, when X_3 is unconditionally dependent on X_1 ($X_1 \not\perp X_3$), as depicted in the SCM 12a and DAG 12b, the BNF express their joint probability distribution as $P(X_1, X_3) = P(X_3 \mid X_1)P(X_1)$. Notably, these descriptions demonstrate the clear correspondence between the structural equations illustrated in Section 8.1.3.1 and the CPDs.

Beyond the insights gained from two-node structures, researchers can uncover more nuanced patterns of(in)dependencies from chains, forks, and colliders. These (in)dependencies apply to any data set generated by a causal model with those structures, regardless of the specific functions attached to the SCM (Pearl et al., 2016, p. 36). For instance, applying the BNF to the chain structure depicted in the SCM 13a and DAG 13b allow researchers to represent the joint distribution for the observed variables as $P(X_1, X_2, X_3) = P(X_1)P(X_2 \mid X_1)P(X_3 \mid X_2)$. This expression implies that X_1 and X_3 are unconditionally dependent ($X_1 \not\perp X_3$), but conditionally independent when controlling for X_2 ($X_1 \perp X_3 \mid X_2$). Moreover, in the fork structure shown in the SCM 14a and DAG 14b, researchers can express the joint distribution of the observed variables as $P(X_1, X_2, X_3) = P(X_1 \mid X_2)P(X_2)P(X_3 \mid X_2)$. Similar to the chain structure, this expression allows researchers to further infer that X_1 and X_3 are unconditionally dependent ($X_1 \not\perp X_3$), but conditionally independent when controlling for X_2 ($X_1 \perp X_3 \mid X_2$). Finally, researchers analyzing the collider structure illustrated in the SCM 15a and DAG 15b can express the joint distribution of the observed variables as $P(X_1, X_2, X_3) = P(X_1)P(X_2 \mid X_1, X_3)P(X_3)$. This representation allows researchers to infer that X_1 and X_3 are unconditionally independent ($X_1 \perp X_3$), but conditionally dependent when controlling for X_2 ($X_1 \not\perp X_3 \mid X_2$). The authors Pearl et al. (2016, p. 37, 40, 41) and Neal (2020, p. 25–26) provide the mathematical proofs for these conclusions.

Using these additional probabilistic insights, researchers can revisit the scenario in Section 8.1.2. In this context, applying the BNF to the SCM 17a structure, enables the representation of the joint probability distribution of the observed variables as $P(Y, T, X) = P(Y \mid T, X)P(T \mid X)P(X)$. From this expression, researchers can infer that the outcome Y is unconditionally dependent on the teaching method T ($Y \not\perp T$). This dependency arises from two key structures: a direct causal path from the teaching method T to the outcome Y, represented by the two-connected-nodes structure $T \to Y$ (black path in DAG 17b), and a confounding non-causal path from the teaching method T to the outcome Y through the socio-economic status of the school X, represented by the fork structure $T \leftarrow X \to Y$ (gray path in DAG 17b).

$$
\begin{aligned}
X &:= x \\
T &:= f_T(x, e_T) \\
Y &:= f_Y(T, x, e_Y) \\
e_T &\perp e_X \\
e_T &\perp e_Y \\
e_X &\perp e_Y
\end{aligned}
$$

(a) SCM



(b) Conditioned DAG

Figure 17: Plausible causal structure the scenario outlined in Section 8.1.2.

8.1.3.3. From probability to causality.

The structural approach to Causal inference translates probabilistic insights into actionable strategies seeking to identify the ACE from associational quantities. The approach achieves this by relying on the *modularity assumption*, which posits that intervening on a node alters only the causal mechanism of that node, leaving others unchanged (Neal, 2020, p. 34).

The modularity assumption underpins the concepts of manipulated graphs and Truncated Factorization, which are essential for representing interventions $P(Y_i \mid do(T_i = t))$ within SCMs and DAGs. *Manipulated graphs* simulate physical interventions by removing specific edges from a DAG, while preserving the remaining structure unchanged (Neal, 2020, p. 34). In parallel, *Truncated Factorization* (TF) achieves a similar simulation by removing specific functions from the conceptual model and replacing them with constants, while keeping the rest of the structure unchanged (Pearl, 2010). The probabilistic implications of this factorization are formalized in Equation 25, where S represents the subset of variables X_p directly influenced by the intervention, while an example illustrating these concepts follows below.

$$P(X_1, X_2, \dots, X_P \mid do(S)) = \begin{cases} \prod P(X_p \mid pa(X_p)) & \text{if } p \notin S \\ 1 & \text{otherwise} \end{cases} \tag{25}$$

Using the TF, researchers can define the *backdoor adjustment* to identify the ACE. This adjustment states that if a variable $X_p \in S$ serves as a *sufficient adjustment set* for the effect of X_a on X_b, then the ACE can be identified using Equation 26. The sufficient adjustment set (potentially empty) must block all non-causal paths between X_a and X_b without introducing new paths. If such a set exists, then X_a and X_b are *d-separated* by X_p ($X_a \perp X_b \mid X_p$) (Pearl, 2009), and X_p satisfies the *backdoor criterion* (Neal, 2020, p. 37).

$$P(X_a \mid do(X_b = x)) = \sum_{Xp} P(X_a \mid X_b = x, X_p) P(X_p) \tag{26}$$

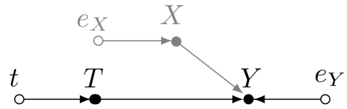
Ultimately, the backdoor adjustment enables researchers to express the ACE as:

$$\begin{aligned} \tau &= E[X_a \mid do(X_b = 1)] - E[X_a \mid do(X_b = 2)] \\ &= E_{Xp}\left[E[X_a \mid do(X_b = 1), X_p] - E[X_a \mid do(X_b = 2), X_p]\right] \\ &= \sum_{Xp} X_a \cdot P(X_a \mid X_b = 1, X_p) \cdot P(X_p) - \sum_{Xp} X_a \cdot P(X_a \mid X_b = 2, X_p) \cdot P(X_p) \end{aligned} \tag{27}$$

With these new insights, researchers revisiting the scenario in Section 8.1.3.2 can infer that the socio-economic status of the school, X, satisfies the backdoor criterion, assuming the causal structure depicted by the SCM 17a and DAG 17b is correct. This means that X serves as a sufficient adjustment set, as it effectively blocks all confounding non-causal paths introduced by the fork structure. Nevertheless, since Y remains dependent on T even after conditioning ($Y \not\perp T \mid X$), this dependency can only be attributed to the direct causal effect $T \rightarrow Y$. Notably, for the purpose of identification, the conditioned DAG 17b is equivalent to the manipulated DAG 18b, because X satisfies the backdoor criterion.

$$
\begin{aligned}
X &:= f_X(e_X) \\
T &:= t \\
Y &:= f_Y(t, X, e_Y) \\
e_T &\perp e_X \\
e_T &\perp e_Y \\
e_X &\perp e_Y
\end{aligned}
$$

(a) SCM



(b) Manipulated DAG

Figure 18: Plausible causal structure the scenario outlined in Section 8.1.3.2.

Researchers can then apply the *backdoor adjustment* to identify the ACE of T on Y. They achieve this by first identifying the CACE of T on Y by conditioning on X, and then marginalizing this effect over X to obtain the ACE. This process is expressed in Equation 28 (see Section 8.1.2).

$$
\begin{aligned}
\tau &= E[Y_i \mid do(T_i = 1)] - E[Y_i \mid do(T_i = 2)] \\
&= E_X \left[E[Y_i \mid T_i = 1, X] - E[Y_i \mid T_i = 2, X] \right] \\
&= \sum_X Y_i \cdot P(Y_i \mid T_i = 1, X) \cdot P(X) - \sum_X Y_i \cdot P(Y_i \mid T_i = 2, X) \cdot P(X)
\end{aligned}
\tag{28}
$$

8.1.3.4. The estimation process.

Ultimately, researchers can use Bayesian inference methods to estimate the ACE. The approach begins by defining two probability distributions: the likelihood of the data, $P(X_1, X_2, \ldots, X_P \mid \theta)$, and the prior distribution, $P(\theta)$ (Everitt and Skrondal, 2010), where X_P represents a random variable, and θ represents a one-dimensional parameter space for simplicity. After observing empirical data, researchers can update the priors to posterior distributions using Bayes' rule in Equation 29:

$$
P(\theta \mid X_1, X_2, \ldots, X_P) = \frac{P(X_1, X_2, \ldots, X_P \mid \theta) \cdot P(\theta)}{P(X_1, X_2, \ldots, X_P)} \tag{29}
$$

Given that the denominator on the right-hand side of Equation 29 serves as a normalizing constant independent of the parameter θ, researchers can simplify the posterior updating process into three steps. First, they integrate new empirical data through the likelihood. Second, they update the parameters' priors to a posterior distribution according to Equation 30. Ultimately, they normalize these results to obtain a valid probability distribution.

$$P(\theta \mid X_1, X_2, \dots, X_P) \propto P(X_1, X_2, \dots, X_P \mid \theta) \cdot P(\theta) \tag{30}$$

Temporarily setting aside the definition of prior distributions $P(\theta)$, note that the posterior updating process depends heavily on the assumptions underlying the likelihood of the data. However, as the number of random variables, P, increases, this joint distribution quickly becomes intractable (Neal, 2020). This intractability is evident from Equation 31, where the likelihood distribution is expressed by multiple chained CPDs.

$$P(X_1, X_2, \dots, X_P \mid \theta) = P(X_1 \mid \theta) \prod_{p=2}^{P} P(X_i \mid X_{i-1}, \dots, X_1, \theta) \tag{31}$$

Nevertheless, researchers can manage the complexity of the likelihood by assuming specific local (in)dependencies among variables. SCMs and DAGs provide a formal framework to represent these assumptions, as detailed in Section 8.1.3.2. These assumptions improve model tractability and simplify the estimation process by enabling the derivation of the BNF of the likelihood (Equation 32). With this simplified structure, any probabilistic programming language can model the system and compute the parameter's posterior distribution using Equation 29.

$$P(X_1, X_2, \dots, X_P \mid \theta) = P(X_1 \mid \theta) \prod_{p=2}^{P} P(X_i \mid pa(X_i), \theta) \tag{32}$$

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